



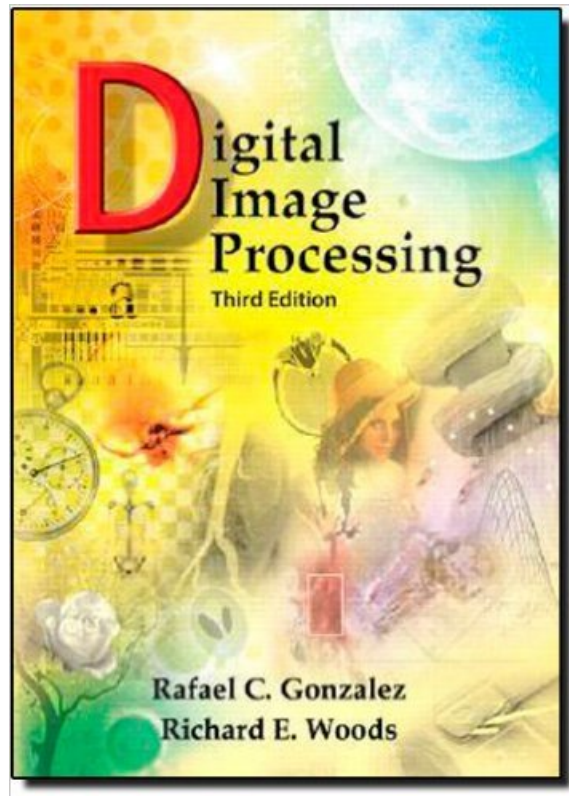
CSE 427

Digital Image Processing

Reference Book

➤ Digital Image Processing (3rd Edition)

Gonzalez and Woods



Goals

- To tell you what you can do with digital images
- To show you that doing research in image processing is fun and exciting
- To demonstrate that image processing is based on strong mathematical principles, applied to digital images via numerical schemes
- To show you that you can solve typical image processing tasks on your own

Image



Image

- A visual representation of something



Digital Image



Digital Image

- A visual representation of something captured by a digital medium

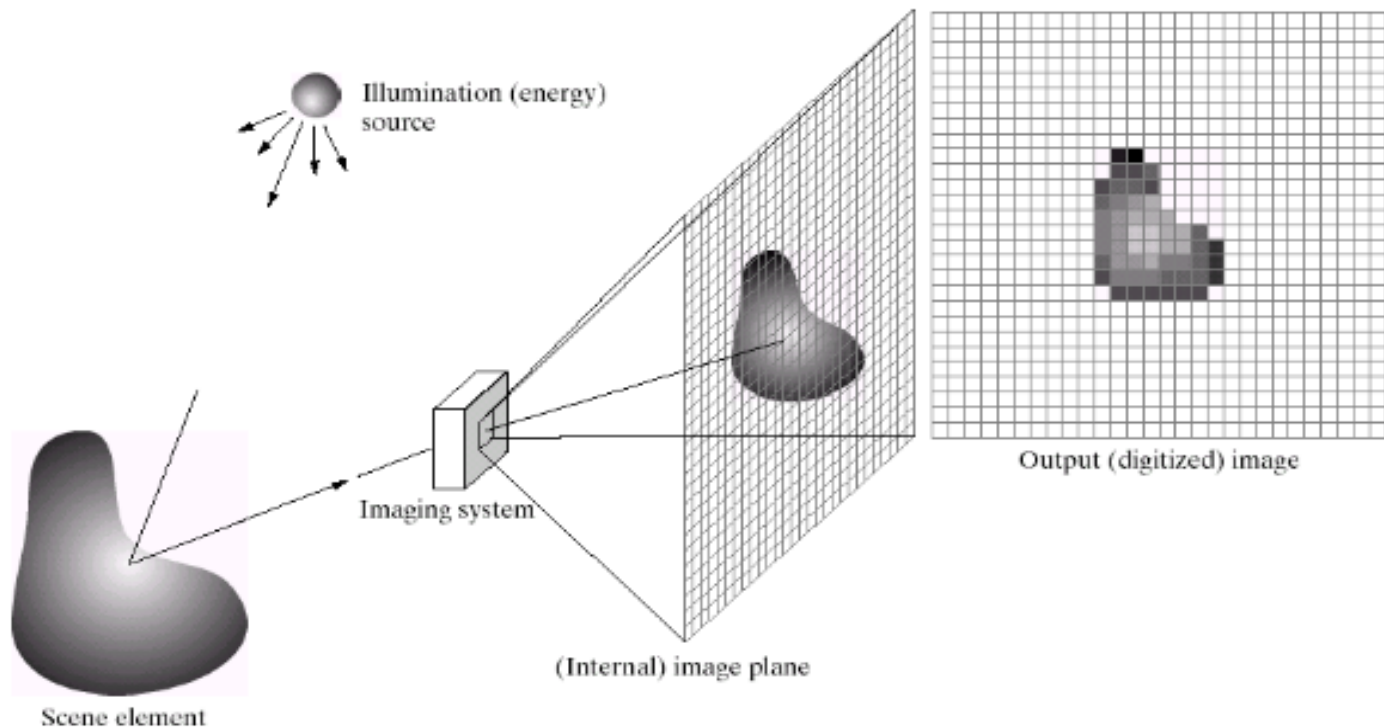


Digital Medium / Image Sensors



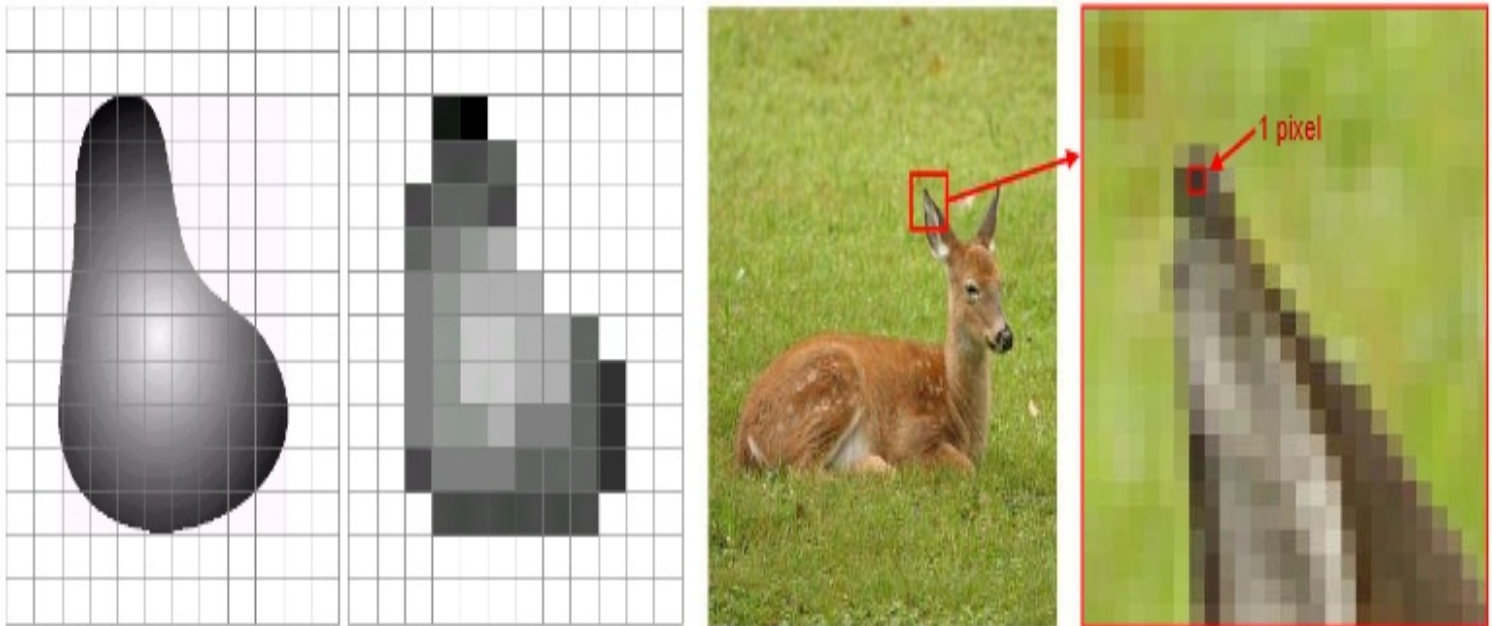
What is a Digital Image?

A digital image is a representation of a two-dimensional image as a finite set of digital values, called picture elements or pixels.



What is a Digital Image?

- Pixel values typically represent gray levels, colors.
- Digital image is an approximation of a real scene.



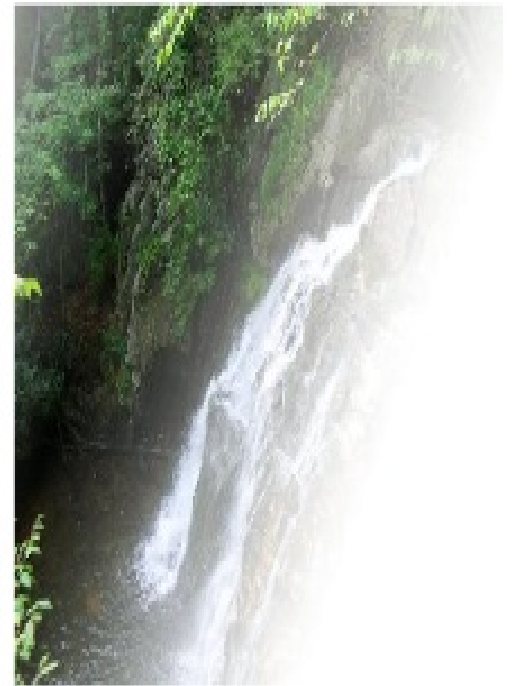
What is a Digital Image?

Common image formats include:

- 1 sample per point (B&W or Grayscale)
- 3 samples per point (Red, Green, and Blue)
- 4 samples per point (Red, Green, Blue, and “Alpha”).

Alpha is for transparency (0.0 fully transparent and 1.0 full opaque (not Transparent)).

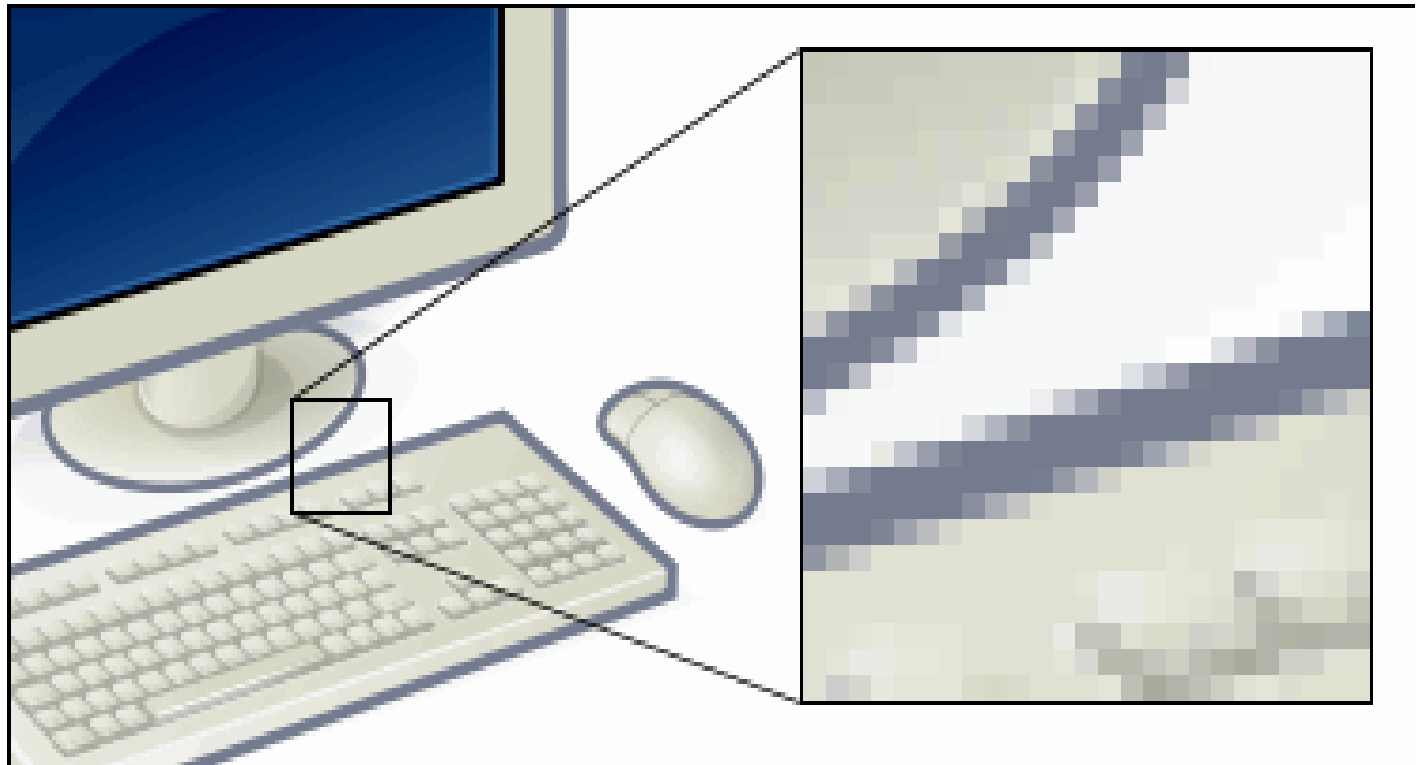
What is a Digital Image?



What is a Digital Image?

- A fixed number of rows and columns of pixels.
- The smallest individual element in an image is pixel.
- In **digital imaging** a pixel, pel, or picture element is a physical point in a image.

What is a Digital Image?



Digital Image



What is a Digital Image?

- Each pixel is a **sample** of an original image.
- More samples typically provide more accurate representations of the original.
- The **intensity** of each pixel is variable.

What is a Digital Image?

- In color imaging systems, a color is typically represented by three or four component intensities such as

For 3 colors:
red, green, and blue, or

For 4 colors:
cyan, magenta, yellow, and black.

What is a Digital Image?

Depending on context, synonyms of pixel as follows:

- pel,
- sample,
- byte,
- bit,
- dot, and
- spot.

Pixels can be used as a unit of measure such as: 2400 pixels per inch, 640 pixels per line, or spaced 10 pixels apart.

What is a Digital Image?

- dots per inch (dpi) and
- pixels per inch (ppi)

For printer devices, dpi is a measure of the printer's density of dot (e.g. ink droplet) placement.

A high-quality photographic image may be printed with 600 ppi on a 1200 dpi inkjet printer.

What is a Digital Image?

Bits Per Pixel (bpp)

The number of distinct colors that can be represented by a pixel depends on the number of bits per pixel (bpp).

A 1 bpp image uses 1-bit for each pixel, so each pixel can be either on or off. Each additional bit doubles the number of colors available, so a 2 bpp image can have 4 colors, and a 3 bpp image can have 8 colors.

What is an Image/Digital Image?

1 bpp, $2^1 = 2$ colors (**monochrome**)

2 bpp, $2^2 = 4$ colors

3 bpp, $2^3 = 8$ colors

...

...

...

8 bpp, $2^8 = 256$ colors

16 bpp, $2^{16} = 65,536$ colors ("**Highcolor**")

24 bpp, $2^{24} = 16,777,216$ colors ("**Truecolor**")

What is an Image/Digital Image?

2^x -> Here x means color depths per pixel.

Depth is normally the sum of the bits allocated to each of the red, green, and blue components.

2^{15} -> Means 5 red, 5 green and 5 blue (divide by 3 channels and always red and blue are equal).

16 bpp means ???

What is an Image/Digital Image?

Highcolor: usually meaning 16 bpp, normally has

- a) Five bits for red and blue each, and
- b) Six bits for green,

As the human eye is more sensitive to errors in green than in the other two primary colors.

What is an Image/Digital Image?

For applications involving **transparency**, the 16 bits may be divided into five bits each of red, green, and blue, with one bit left for transparency.

A 24-bit depth allows 8 bits per component.

On some systems, 32-bit depth is available: this means that each 24-bit pixel has an extra 8 bits to describe its **opacity** (for purposes of combining with another image).

What is an Image/Digital Image?

A **megapixel (MP)** is a million pixels.

The term is used not only for the number of pixels in an image but also

- To express the number of **image sensor** elements of **digital cameras** or
- The number of display elements of **digital displays**.

What is an Image/Digital Image?

In most digital cameras, the sensor array is covered with a patterned color filter having

- Red,
- Green, and
- Blue regions

What is an Image/Digital Image?

Digital image processing focuses on two major tasks

- Improvement of pictorial information for human interpretation
- Processing of image data for storage, transmission and representation for autonomous machine perception.

What is an Image/Digital Image?

- What is the input for image processing or image analysis ???
- What is the output of image processing or image analysis???

Arithmetic Operations

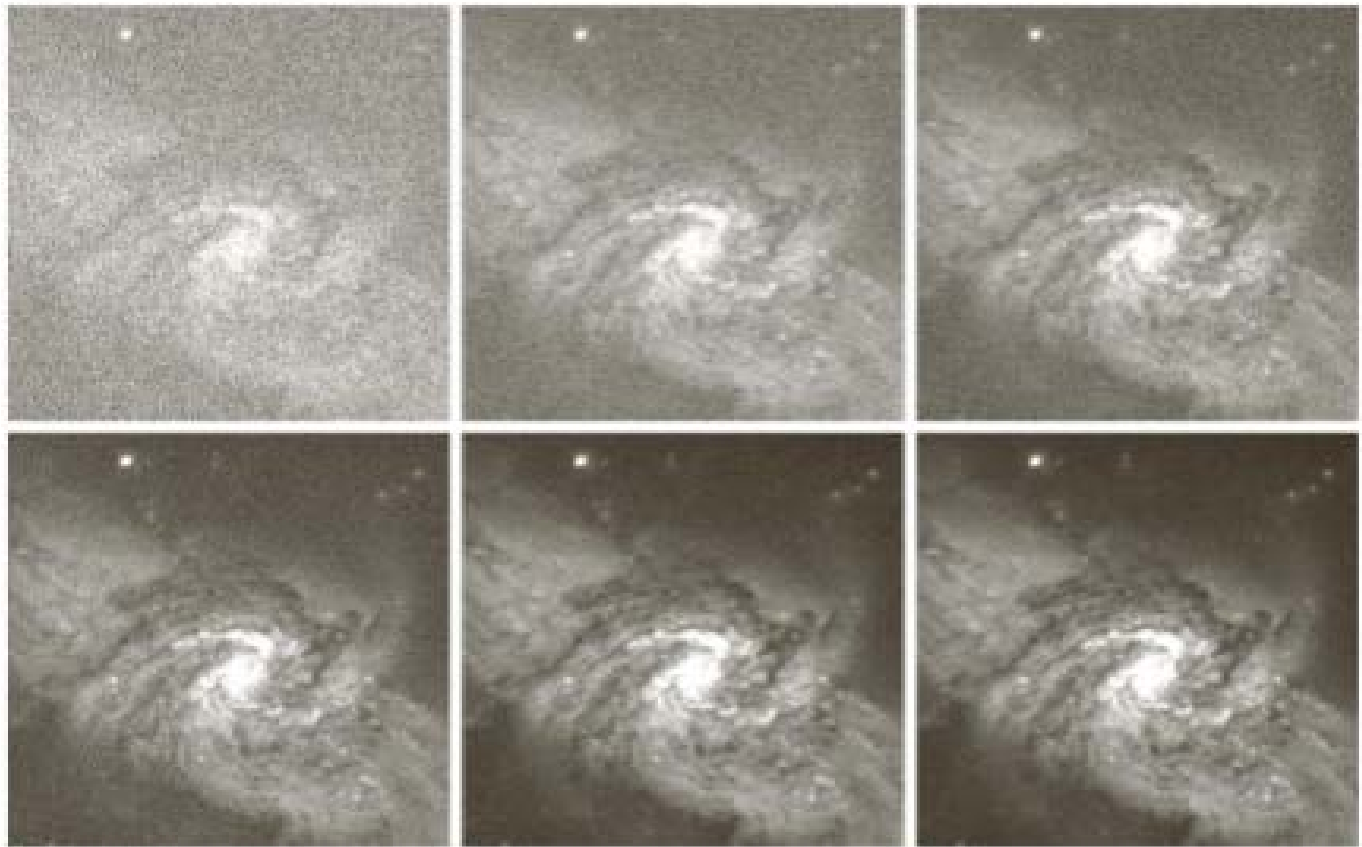
- Arithmetic operations on pixel values

Addition, subtraction, multiply, divide

$$h = f + g \Rightarrow h(i, j) = f(i, j) + g(i, j)$$

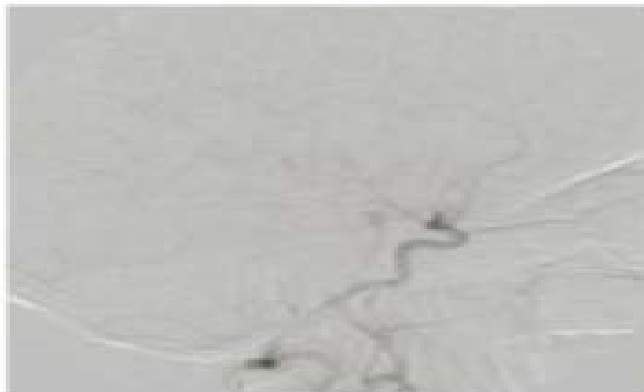
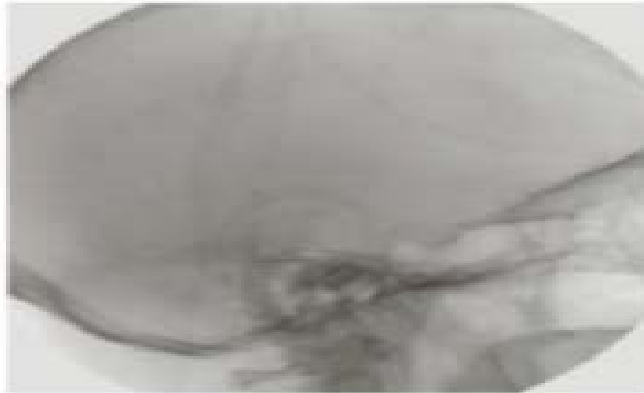
Arithmetic operations: $f + g$

- Averaging (adding) multiple images can reduce



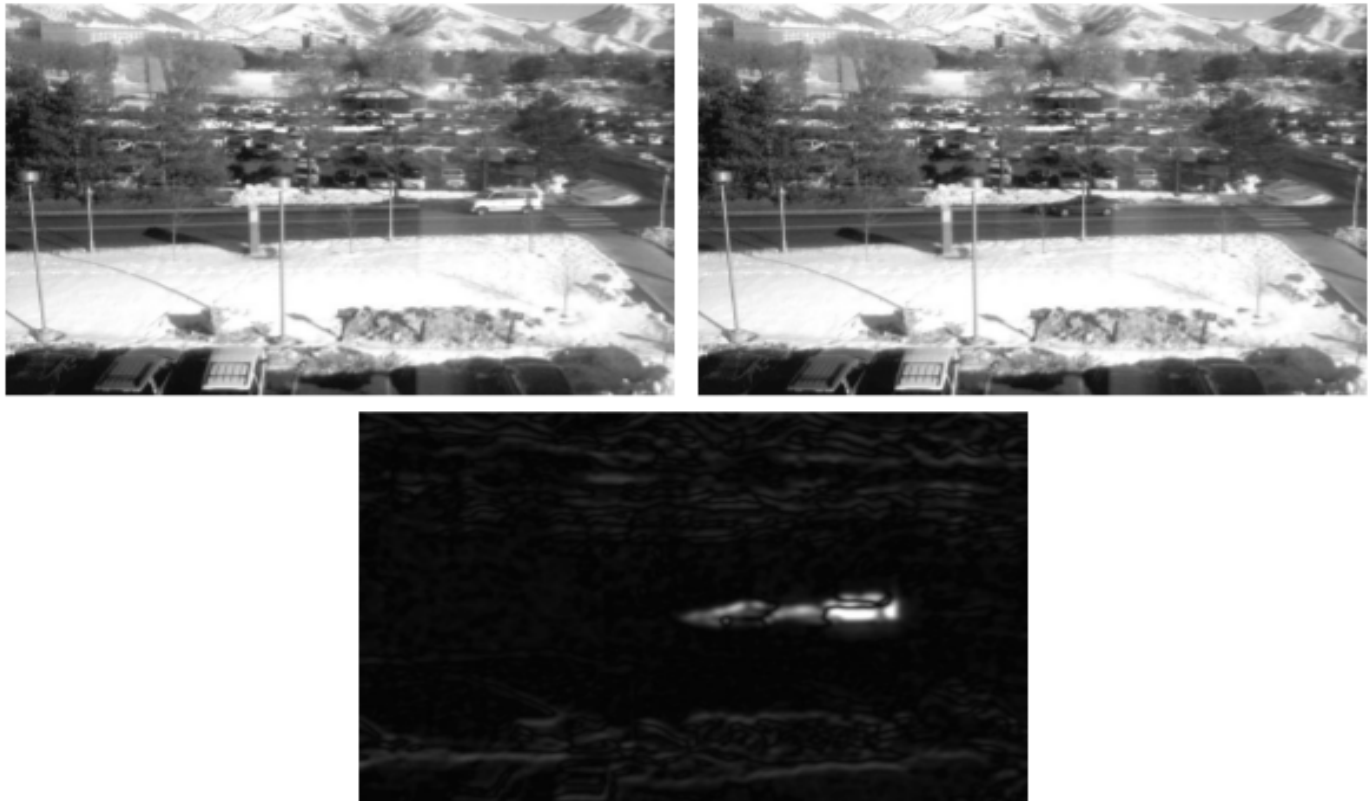
Arithmetic operations: f - g

➤ Digital Subtractive Angiography (DSA)



Arithmetic operations: $f - g$

➤ Image Subtraction (Motion Detection)



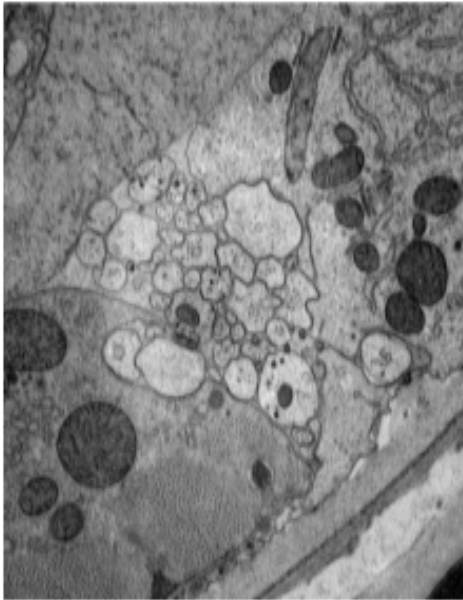
Arithmetic operations: $f \times g$



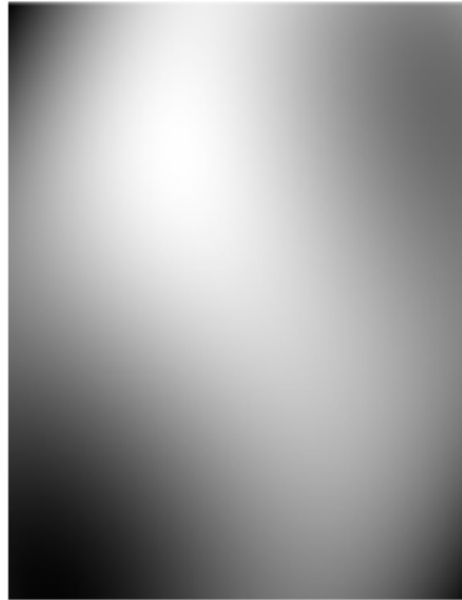
a b c

FIGURE 2.30 (a) Digital dental X-ray image, (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

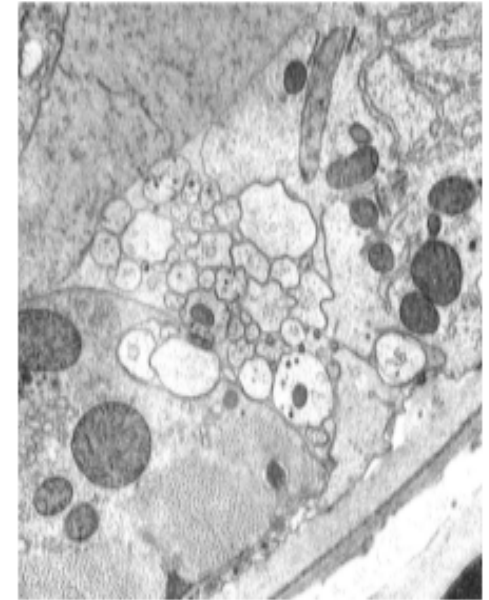
Arithmetic operations: f / g



Captured image

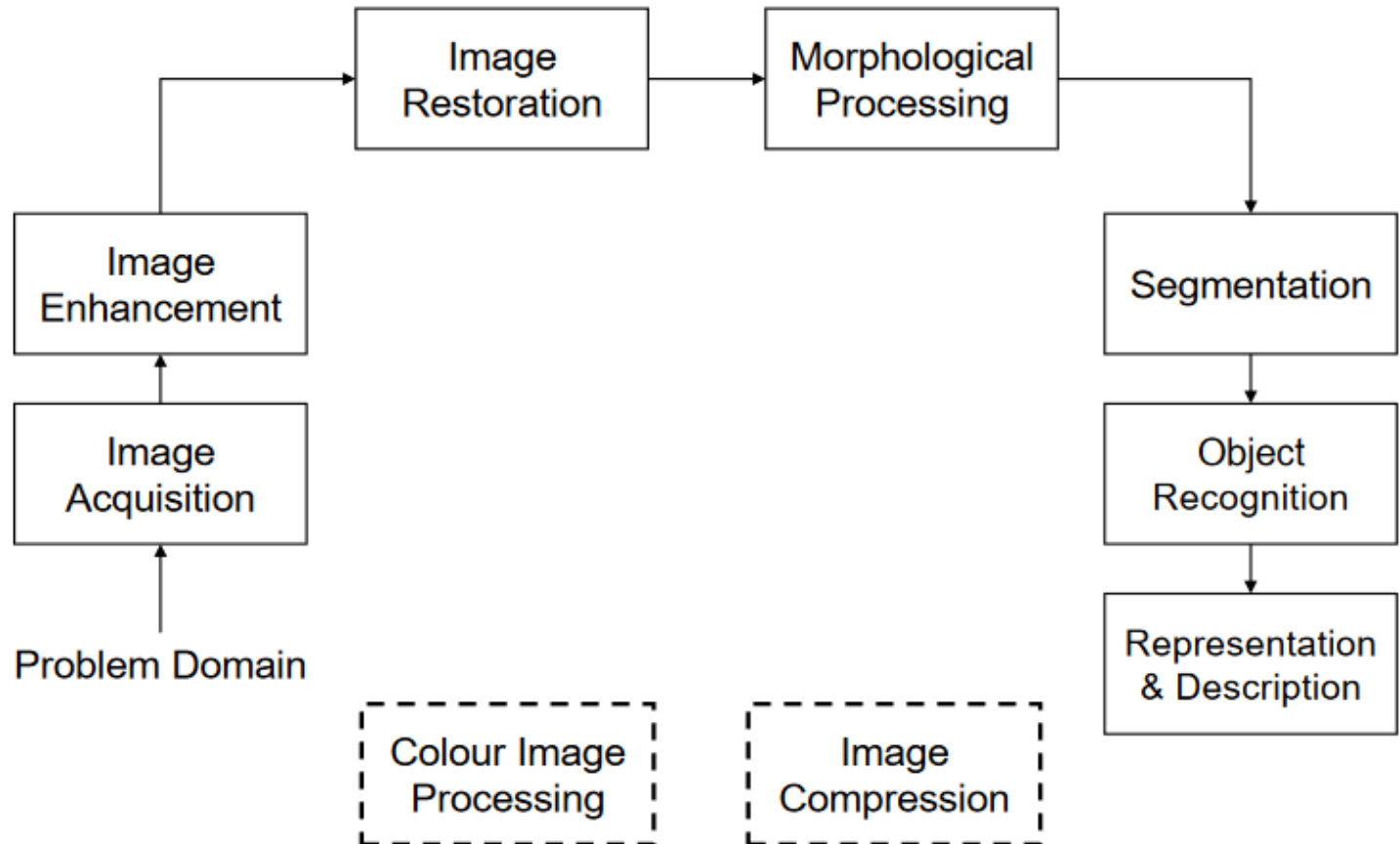


Illumination

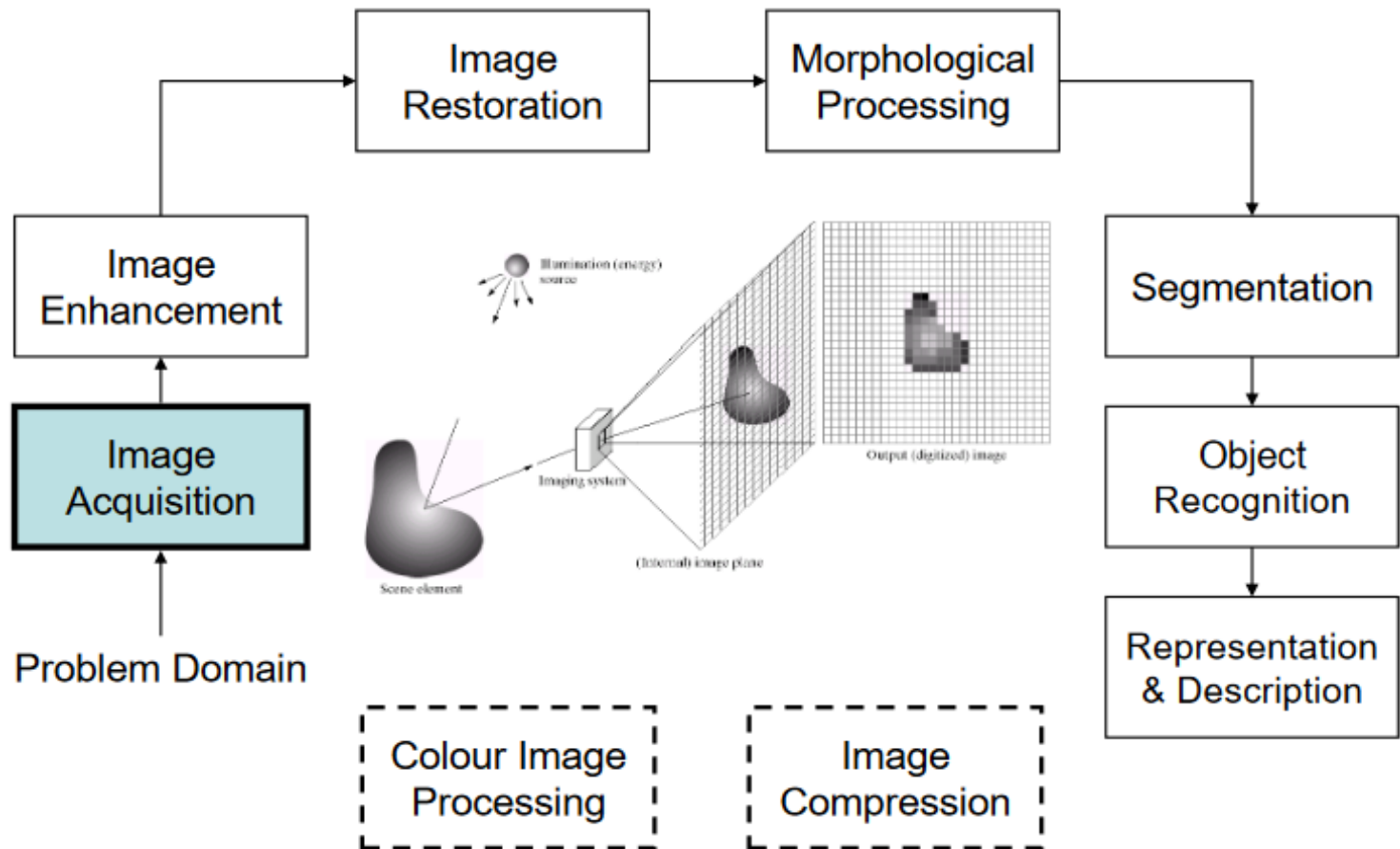


Corrected image

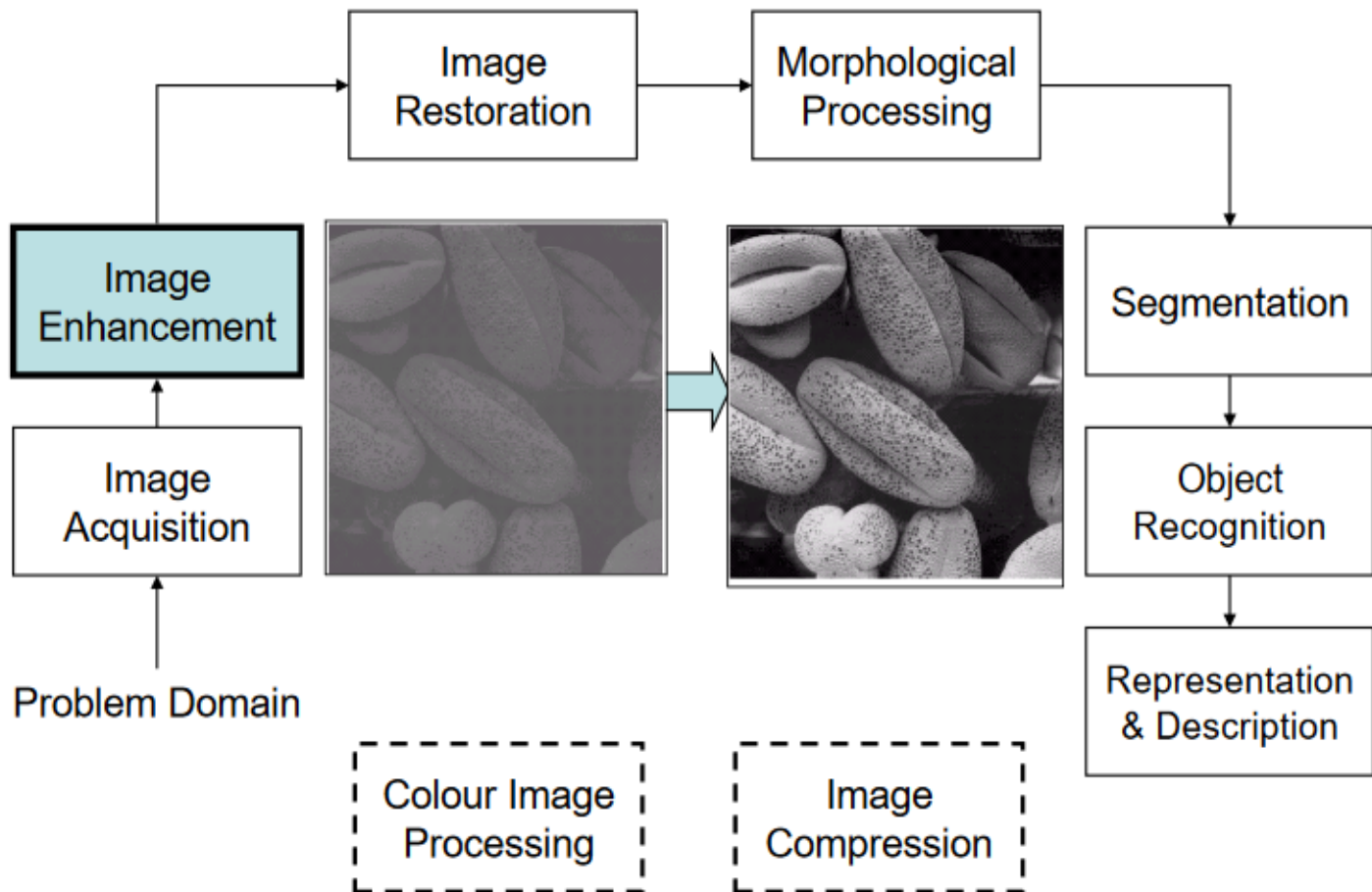
Key Stages in DIP



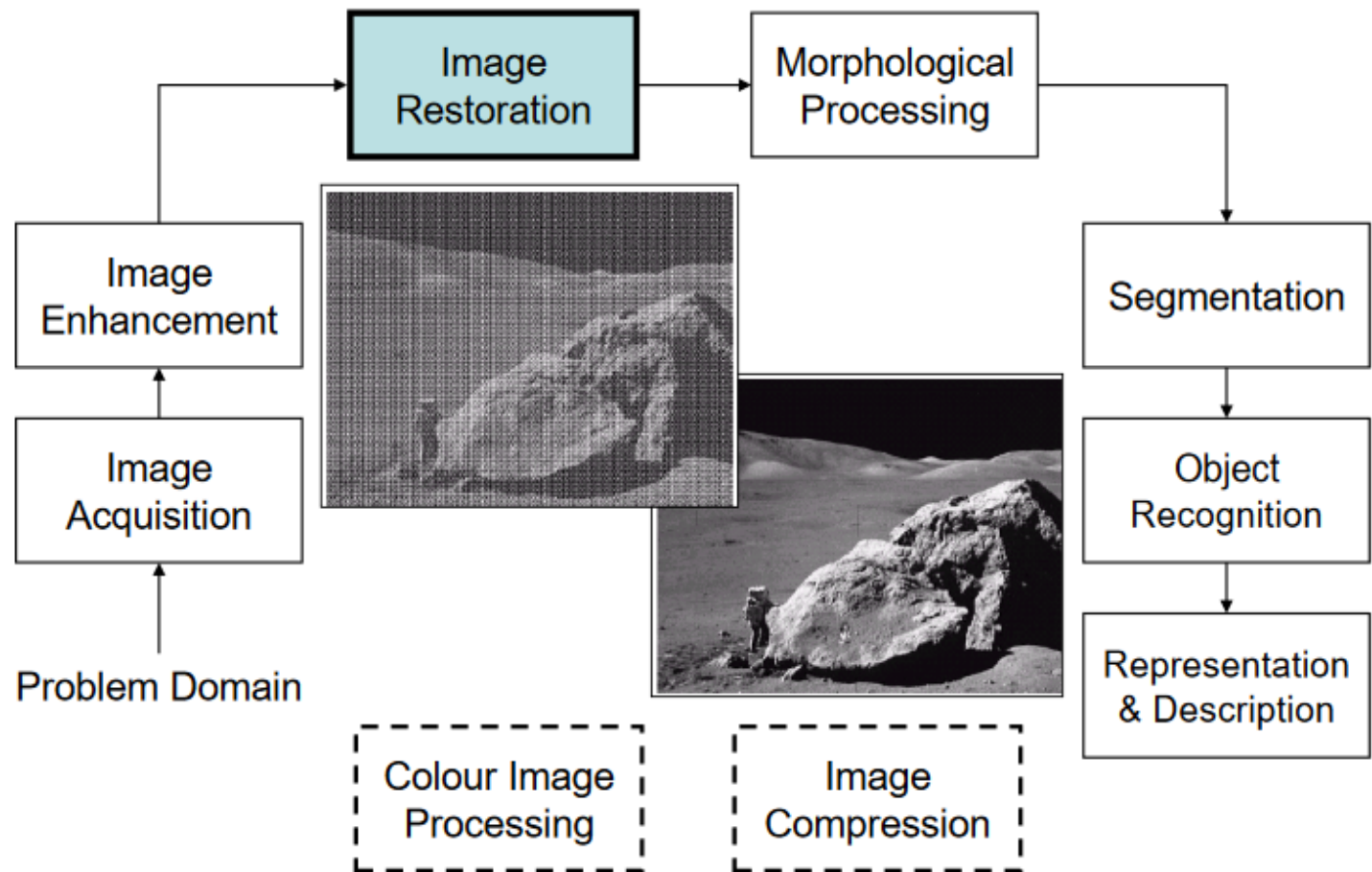
Key Stages in DIP



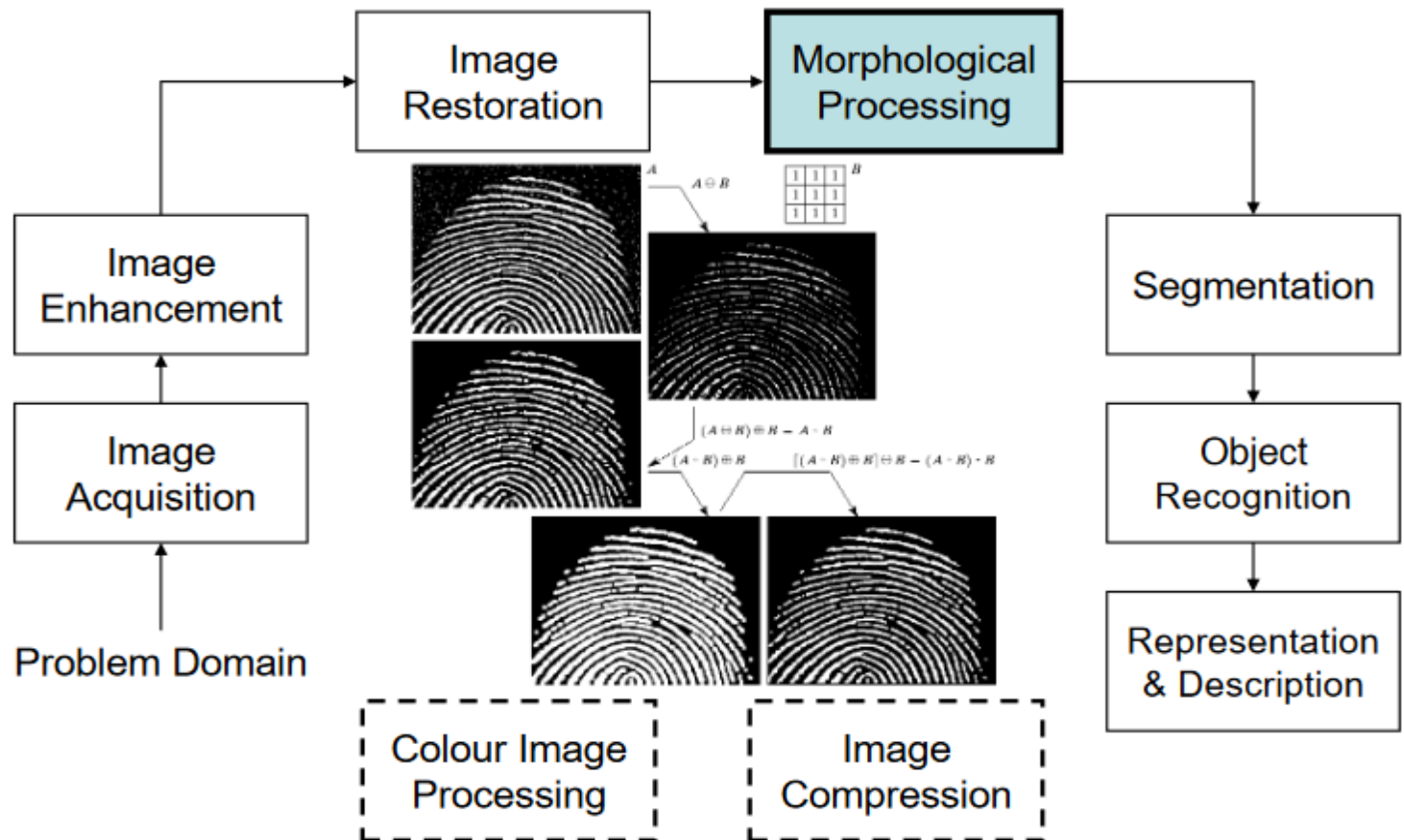
Key Stages in DIP



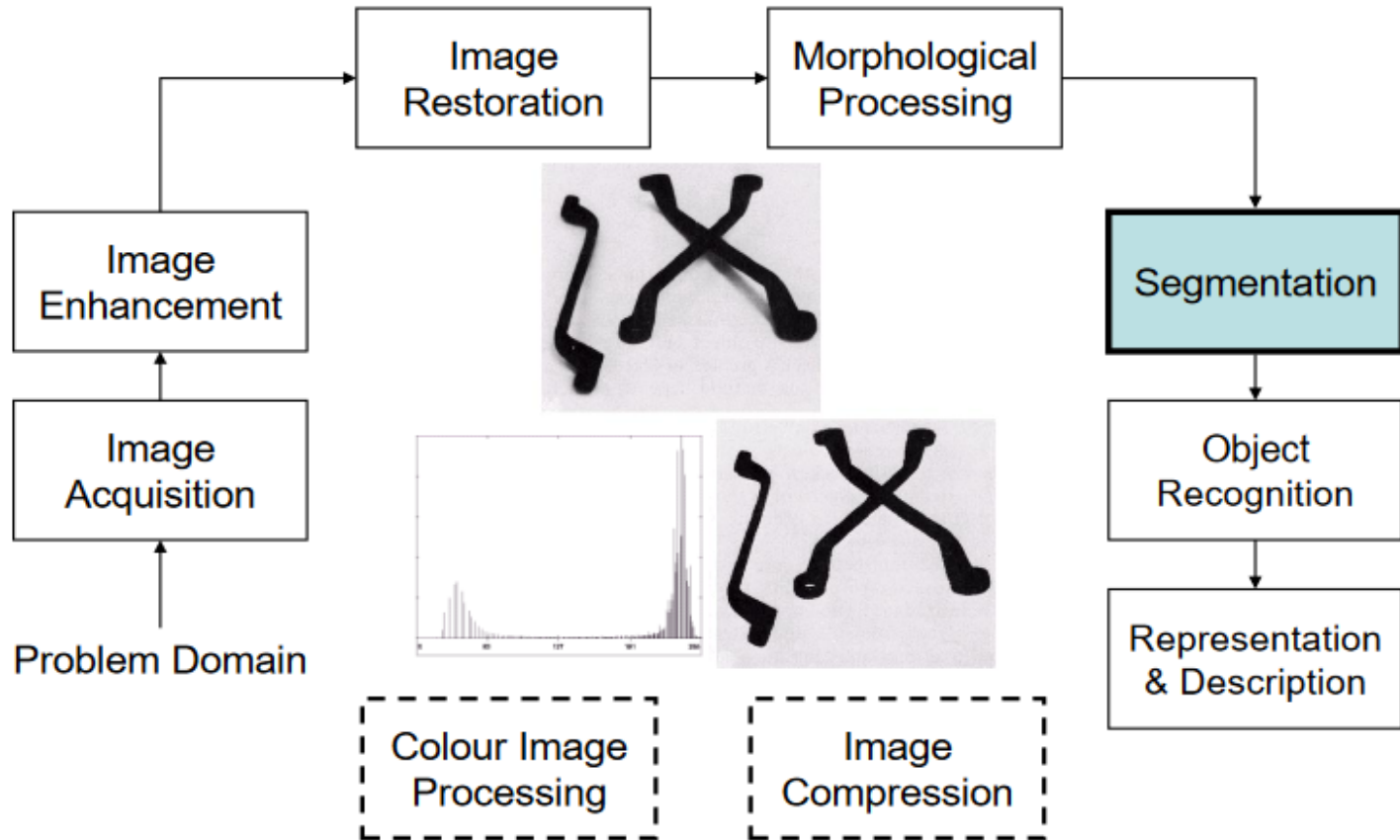
Key Stages in DIP



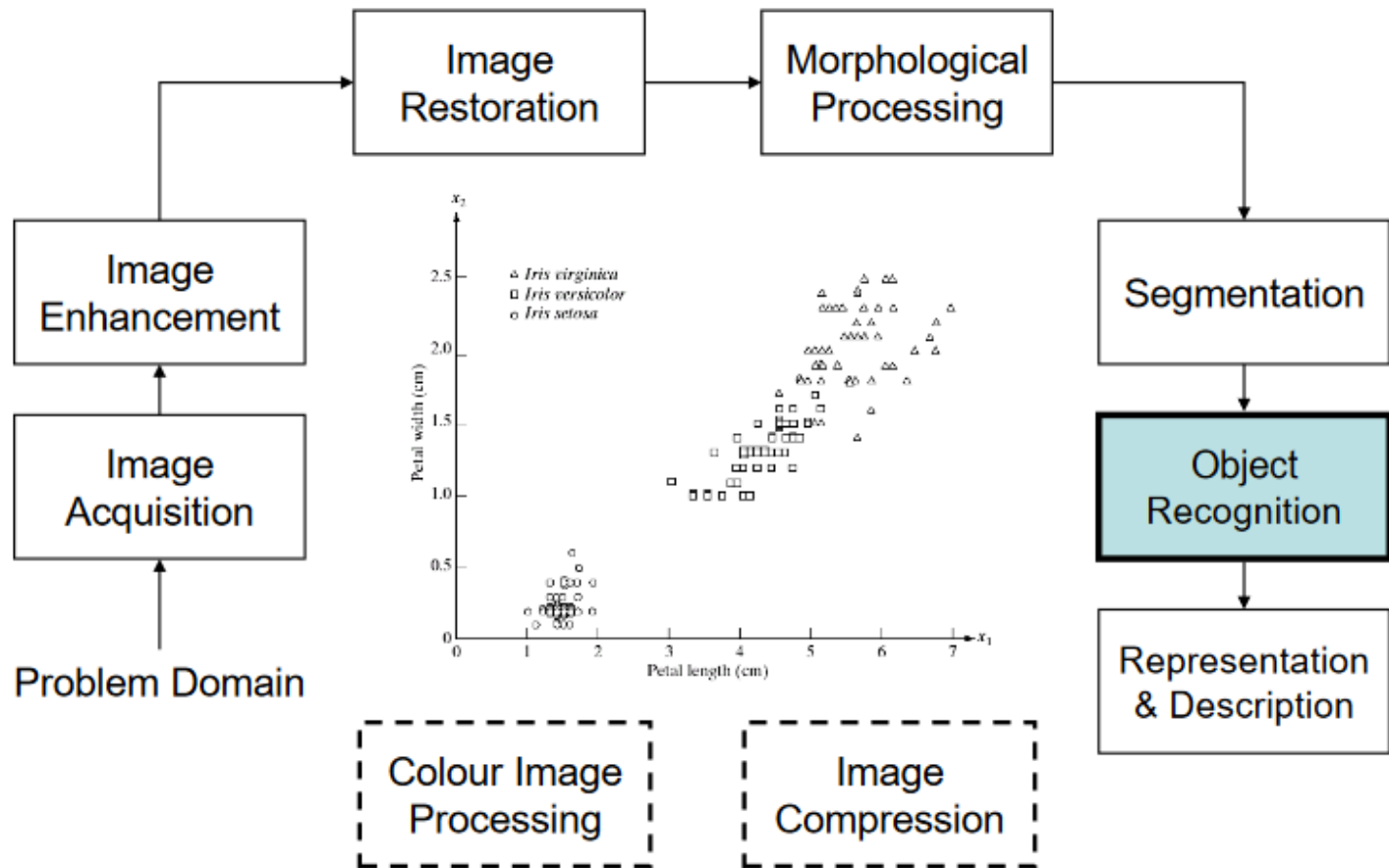
Key Stages in DIP



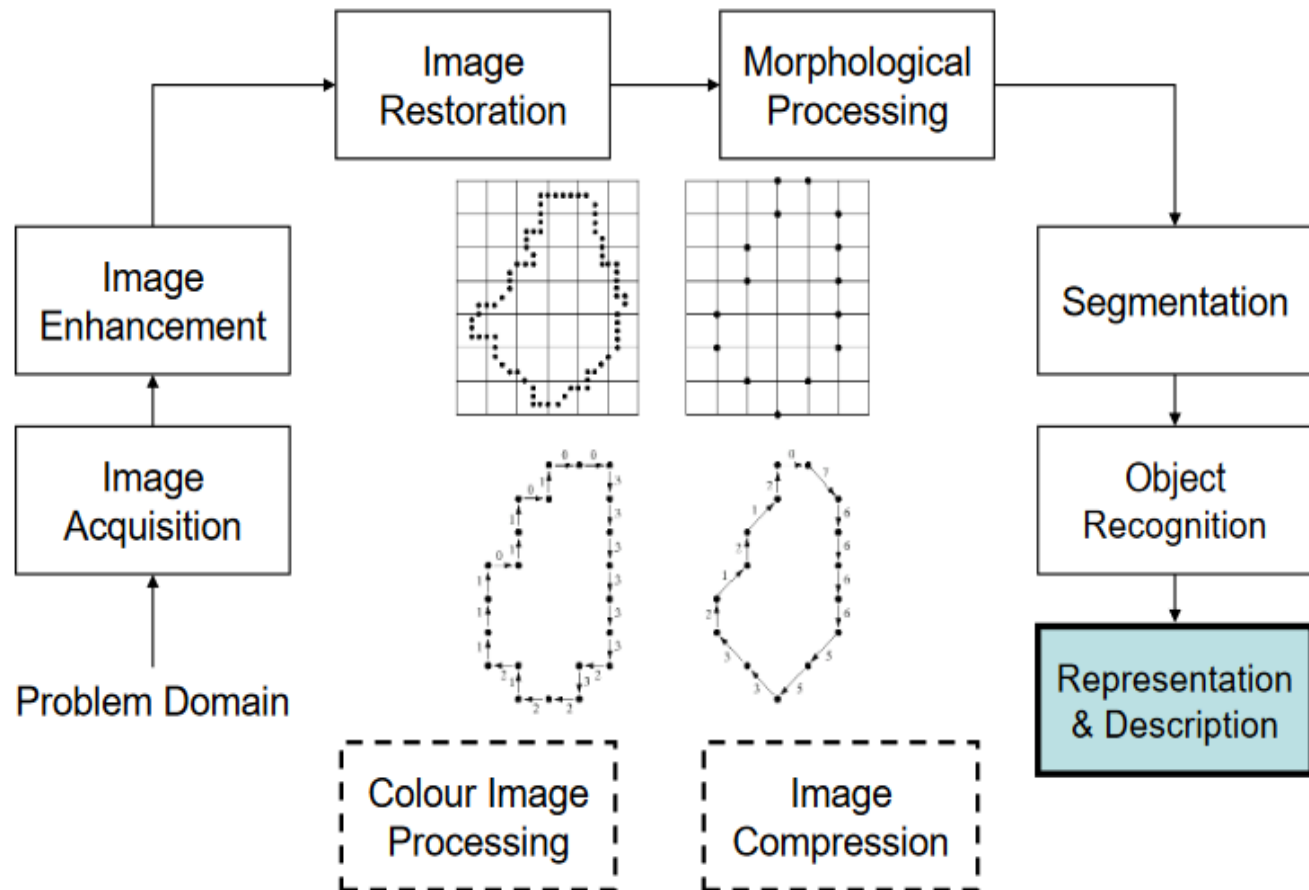
Key Stages in DIP



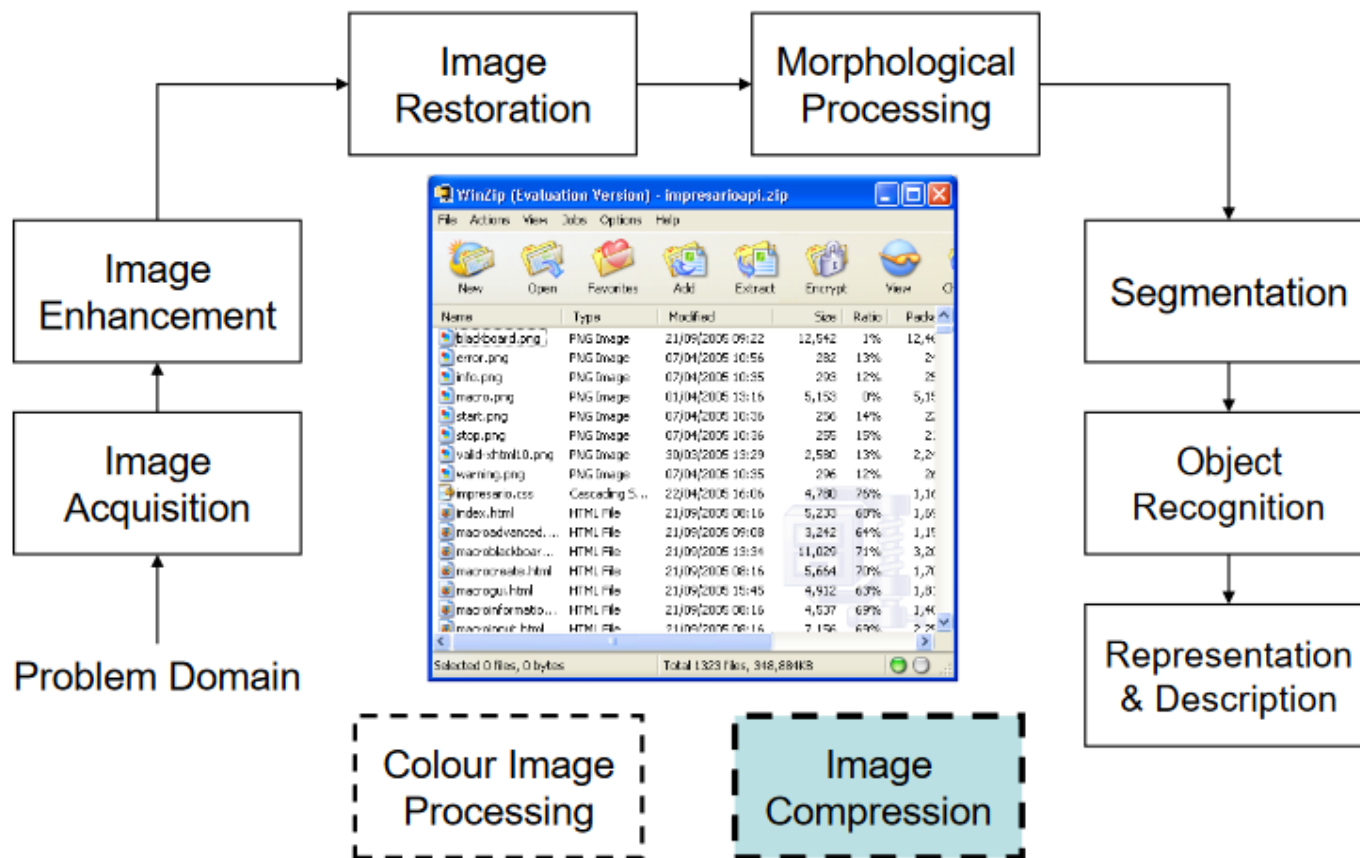
Key Stages in DIP



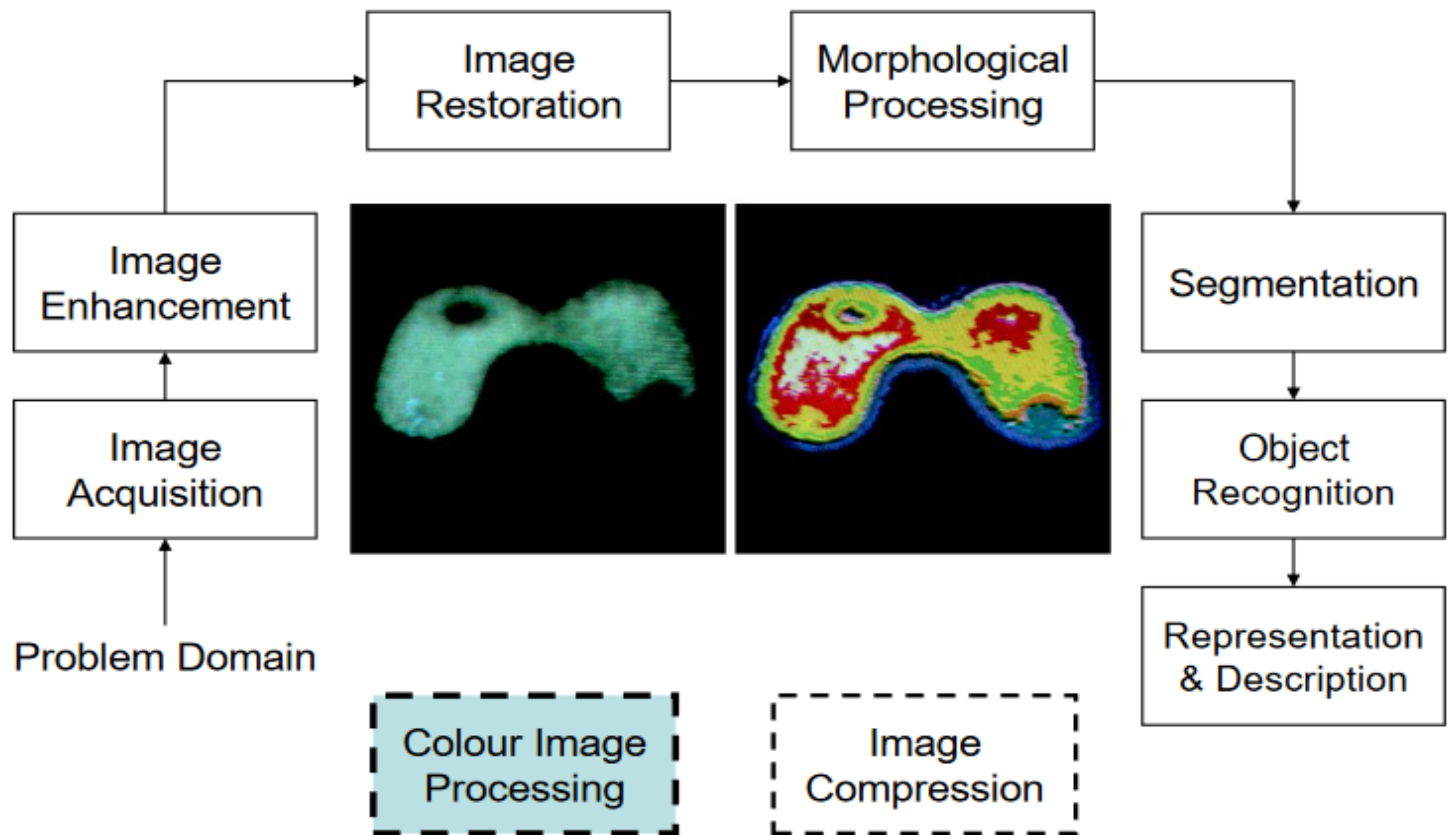
Key Stages in DIP



Key Stages in DIP



Key Stages in DIP



Digital Image Processing

- Processing of *digital images* using *digital devices* (computers)

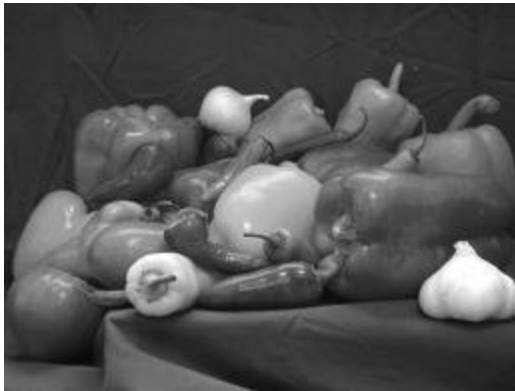


DIP



Scope of DIP

➤ Is it DIP?



Find avg.
gray value



0.67

Scope of DIP

- 3 levels of DIP
 - **Low level:** Input/Output : both are images
 - Example : Enhancement, sharpening, blurring, de-noising
 - **Mid level:** Input: image
 - Output: object, regions
 - Example: Image segmentation
 - **High level:** Input: image
 - Output: class labels
 - Example: Recognition, classification, etc

Why we need DIP

- DIP is not limited to image retrieval (IR)
- Thousands of others areas:
 - Core area
 - Application area

Why we need DIP

- Core area
 - Image enhancement
 - Image de-noising
 - Image segmentation
 - Image & video retrieval
 - Image Security
 - Watermarking and registration
 - Image compression

Why we need DIP

- Application area
 - Security and surveillance
 - Biometric application: face, iris, finger print, palm print recognition . .
 - Medical imaging
 - Automatic scanning and detection: X-ray, CT, MRI, PET, angiogram, ECG, echo, endoscopy,

Why we need DIP

➤ Application area

- Document Classification
 - Binarization, segmentation, character recognition
- Object and shape recognition
- Intelligent transport system
- Environmental monitoring and remote sensing
- and so on, . . .

Digital Image ...

An image can be defined as

➤ a two-dimensional function, $f(x,y)$,
where

a) x and y are spatial (plane) coordinates

b) amplitude of f at any pair of
coordinates (x,y)

is called the intensity of the image at that point.

Digital Image

The term gray level is often used to refer to the intensity of monochrome images.

A monochromatic object or image reflects colors in shades of limited **colors** or **hues**. Images using only shades of grey (with or without black or white) are called **grayscale** or **black-and-white**.

Hue is one of the main properties (called **color appearance parameters**) of a **color**.

Digital Image Processing

Digital image processing refers to processing digital images by means of a digital computer.

- A digital image is composed of a finite number of elements, each of which has a particular location and value.
- These elements are referred to as picture elements, image elements, pels and pixels.
- Pixel is the term most widely used to denote the elements of a digital image.

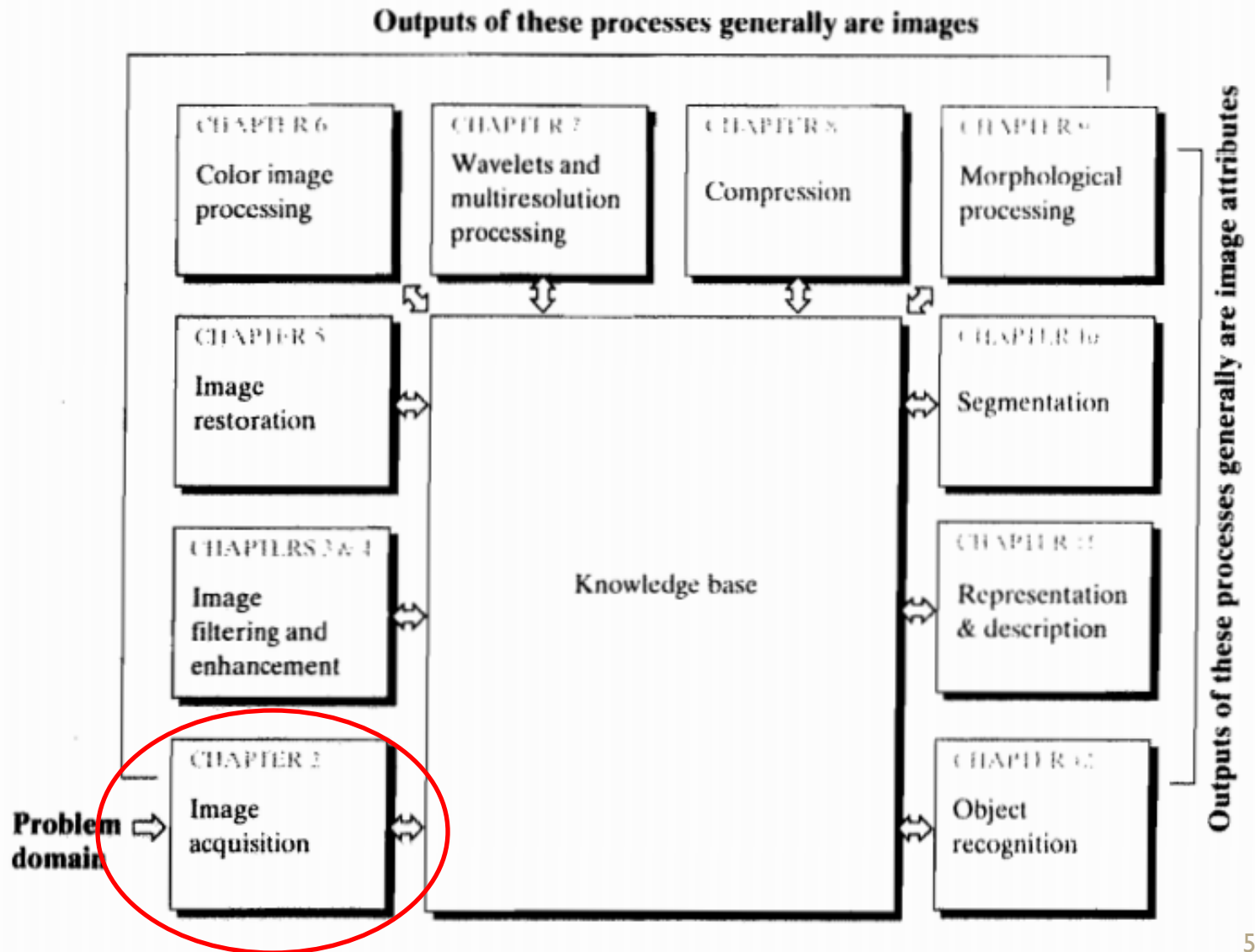


Chapter 2

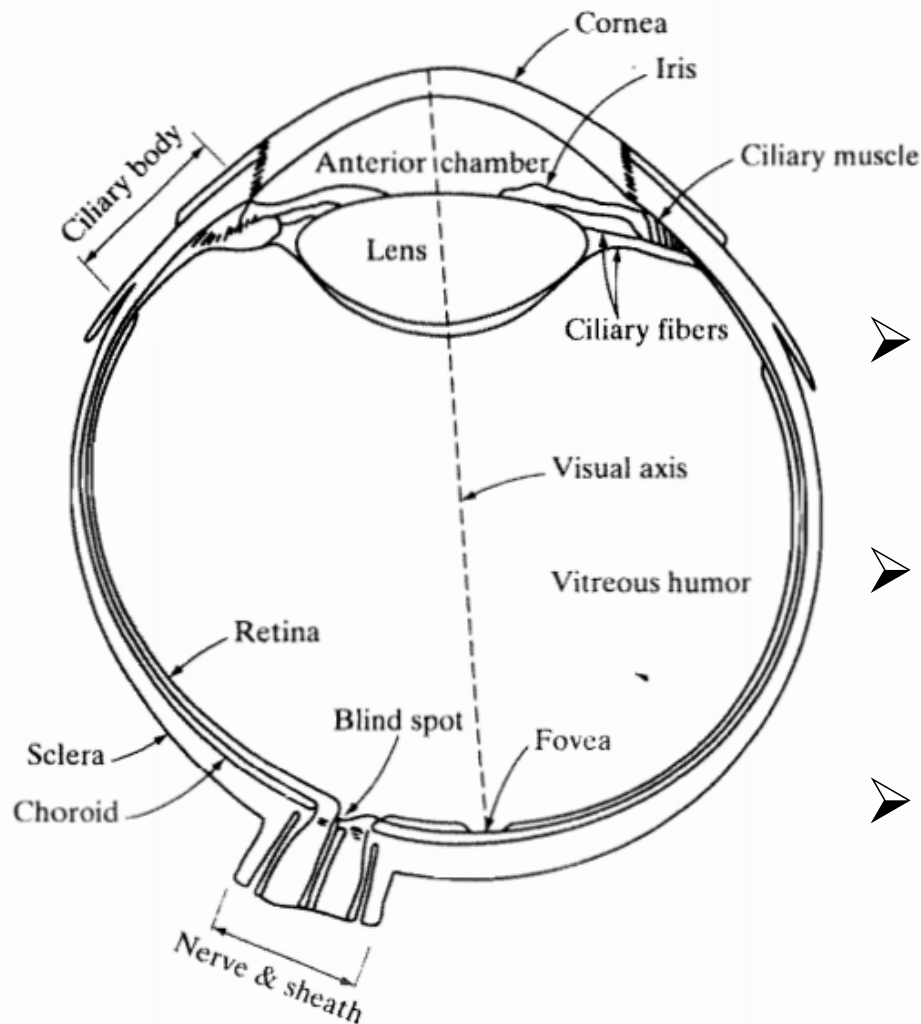
Digital Image Fundamentals

Remember?

FIGURE 1.23
Fundamental steps in digital image processing. The chapter(s) indicated in the boxes is where the material described in the box is discussed.



Structure of Human Eye



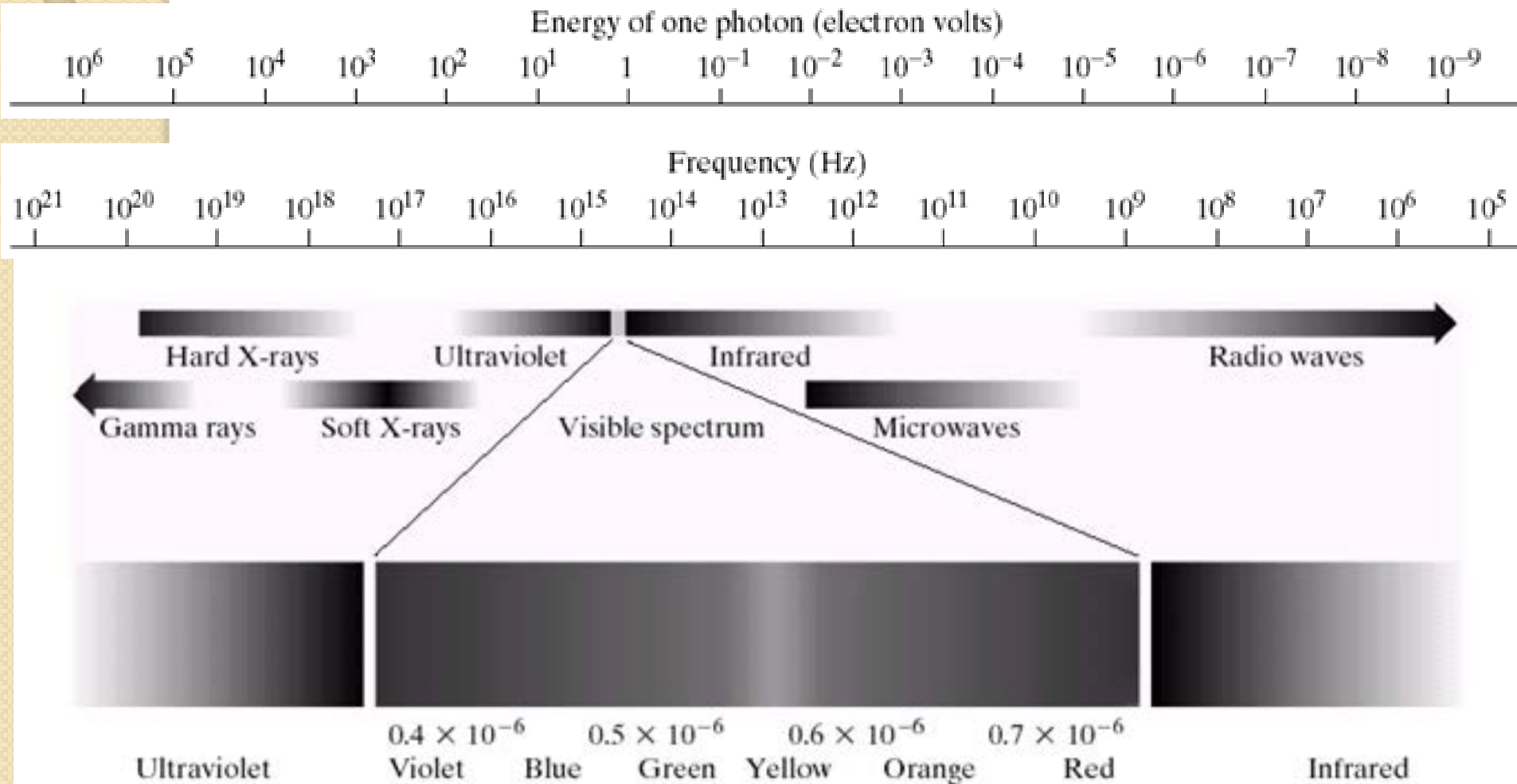
- No need to study for term final.
- You can safely skip section 2.1
- Interested readers are requested to go through it.

EM Spectrum

The **electromagnetic spectrum** is the term used by scientists to describe the entire range of light that exists.

From radio **waves** to gamma rays, most of the light in the universe is, in fact, invisible to us!

EM Spectrum



EM Spectrum

Monochromatic Light

- Light that is void of color is called **monochromatic** light. (A photograph or picture developed or executed in black and white or in varying tones of only one color.)
- The only attribute of monochromatic light is **intensity**. (can be referred to as brightness)
- The intensity of monochromatic light is perceived to vary from black to grays and finally to white.

Monochromatic Light

- The term **gray level** is used to commonly denote monochromatic intensity.
- The range of measured values of monochromatic light from black to white is usually called the **gray scale** and monochromatic images are often referred to as **gray-scale images**.

EM Spectrum

Chromatic Light

- Chromatic light spans the electromagnetic energy spectrum from approximately 0.43 to 0.79 μm

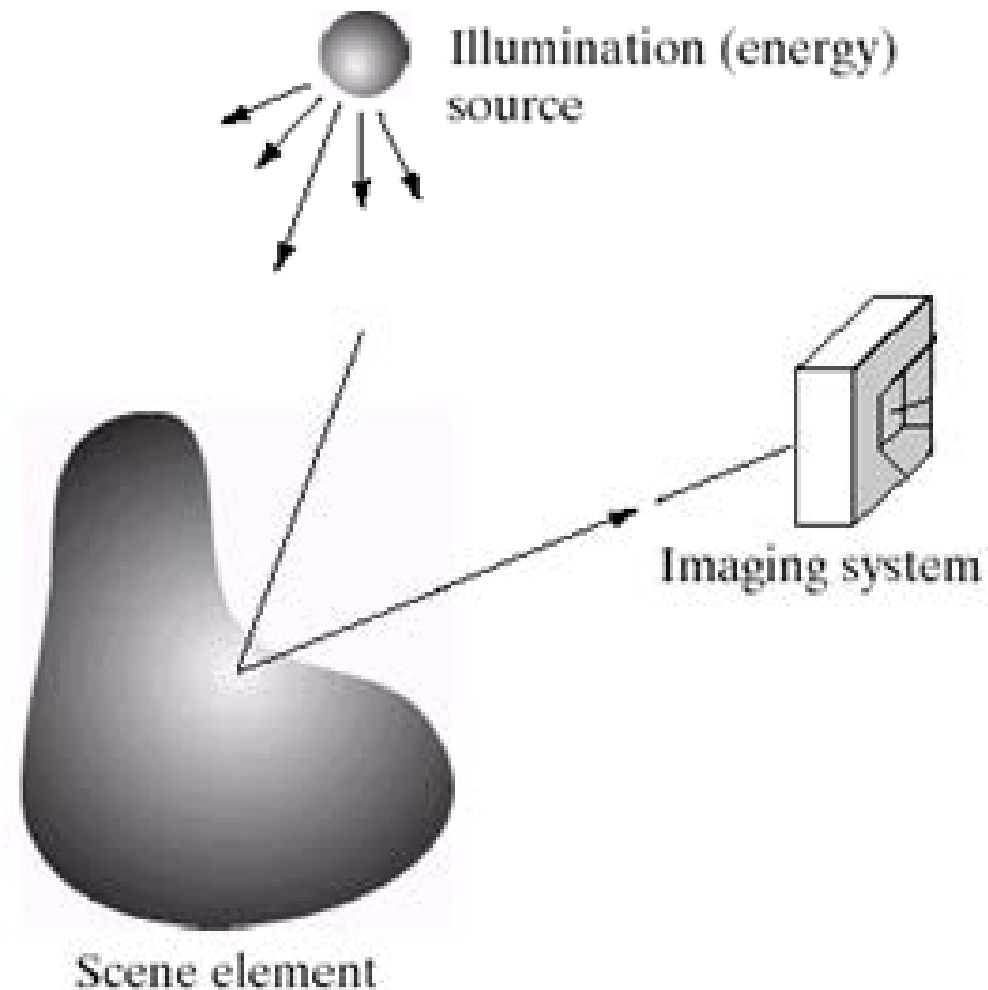


Image Acquisition

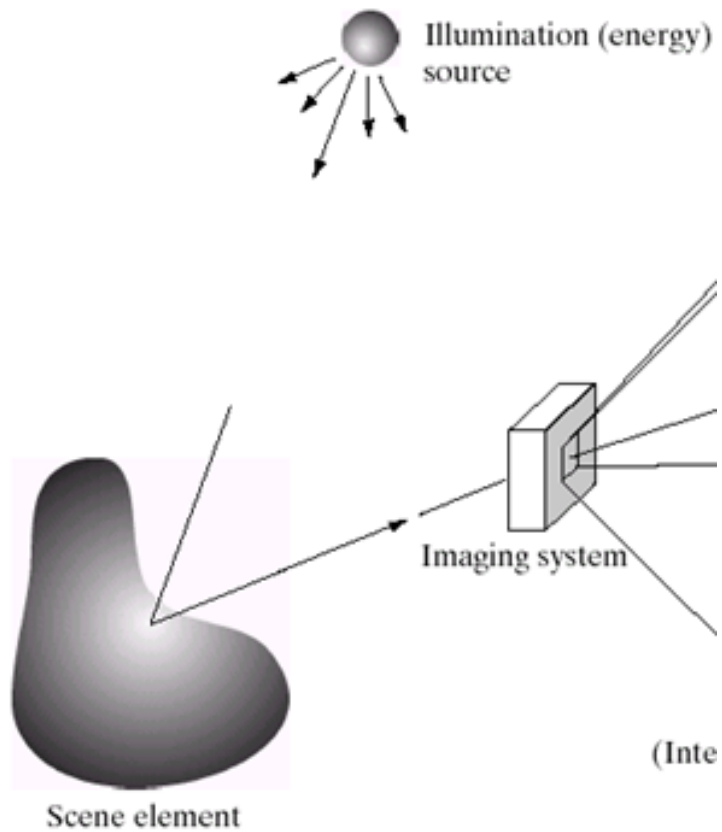
Image Acquisition

Three main elements

- Illumination source
- Scene
- Sensor (imaging system)

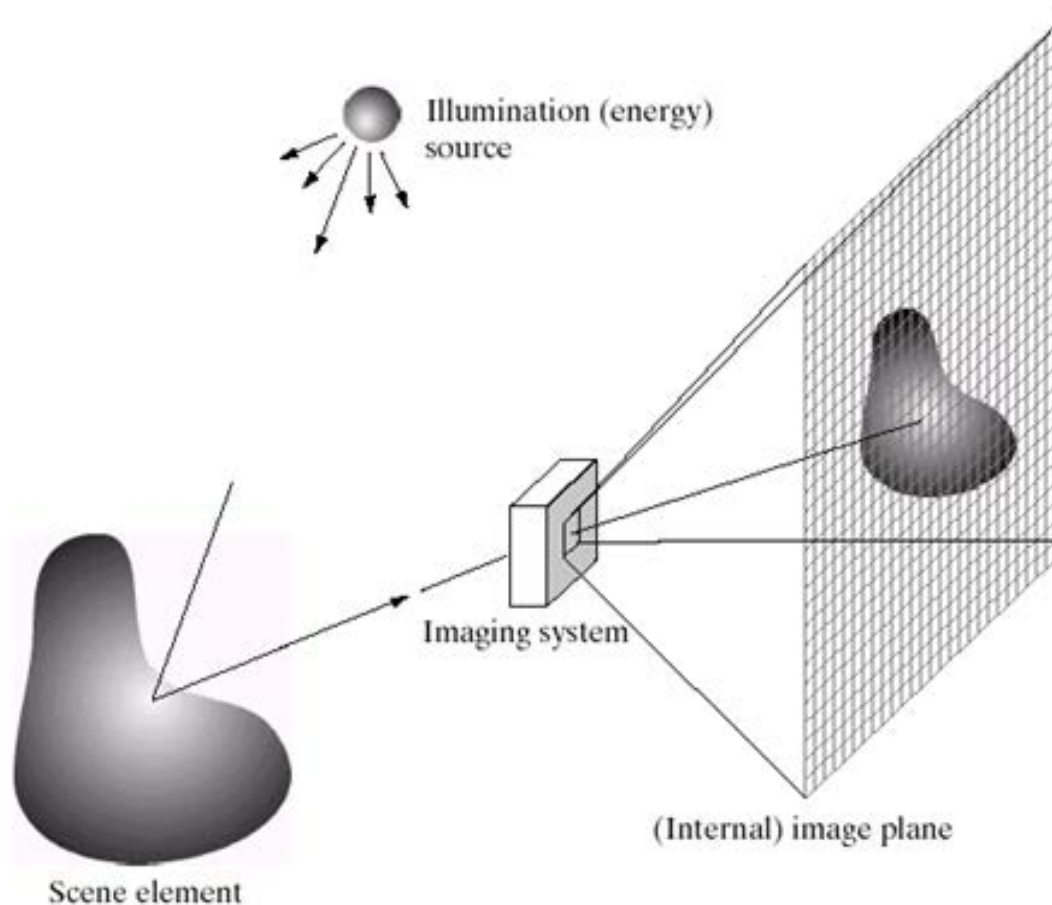


Sensor Array



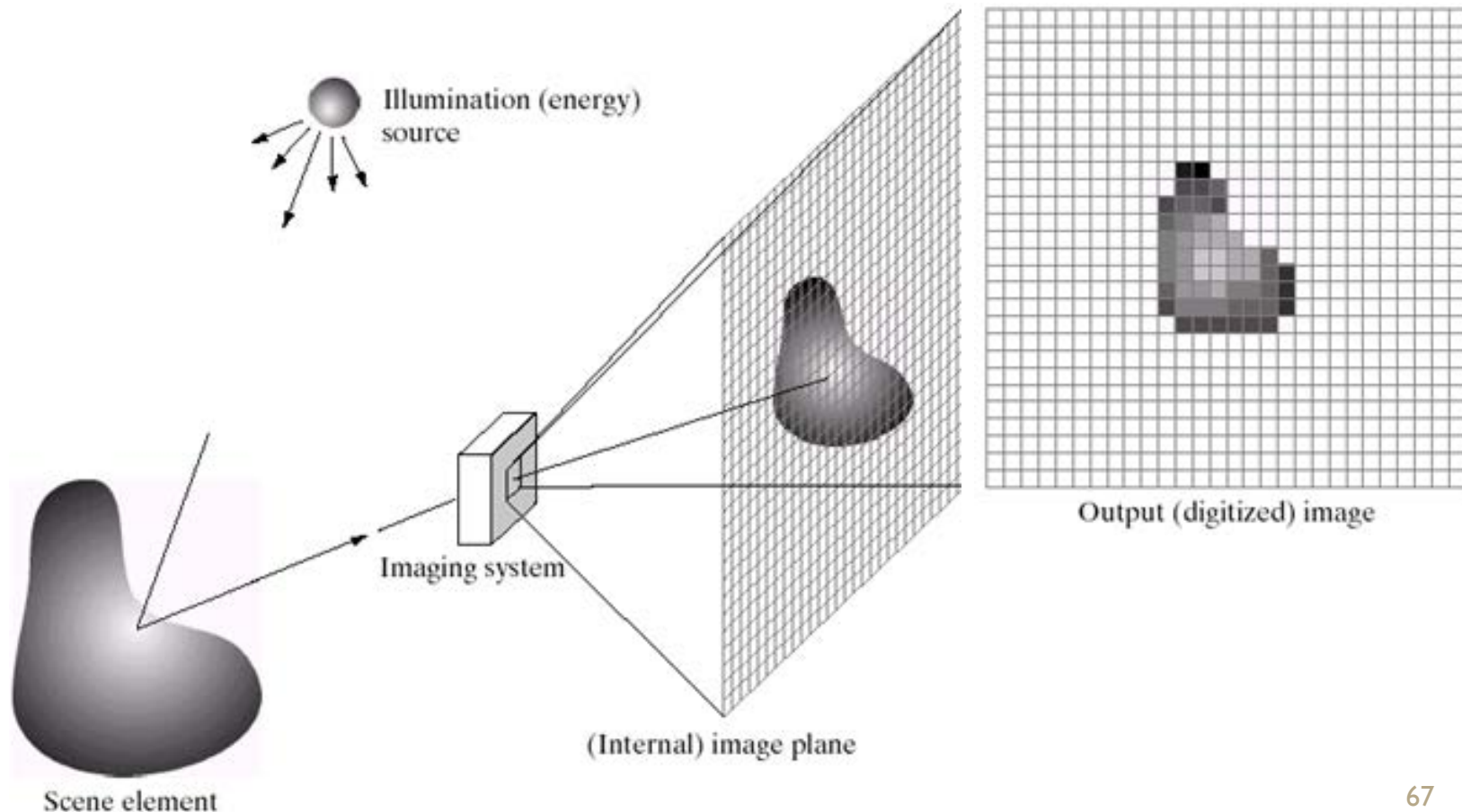
Sensor Array

The sensor array produces outputs proportional to the integral of the light received at each sensor.



Sensor Array

- ° Digital and analog circuitry sweep these outputs and convert them to an analog signal which is then digitized by another section of the imaging system



A Simple Image Formation Model

Recall from previous class:

An image has two basic attributes:

- A location (x, y) : picture element, **pixel**, pel, etc
- A value $f(x, y)$ at pixel (x, y) : gray scale value

$f(x, y)$ represents intensity which is proportional energy radiation.

$$0 < f(x, y) < \infty$$



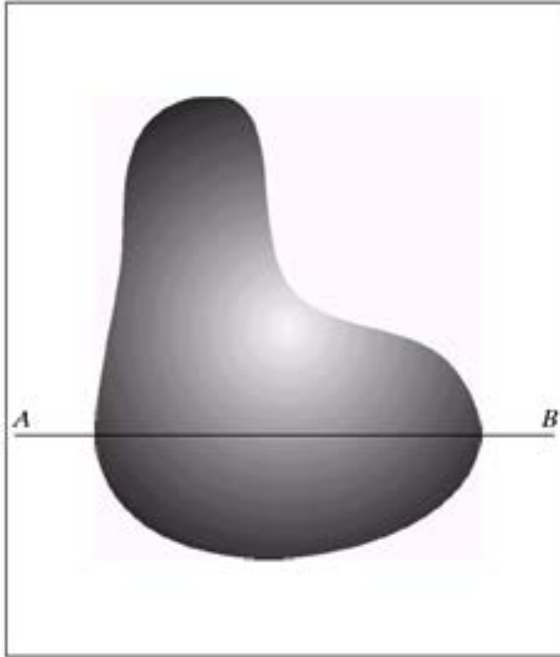
Image Digitization

Image Digitization

Two steps :

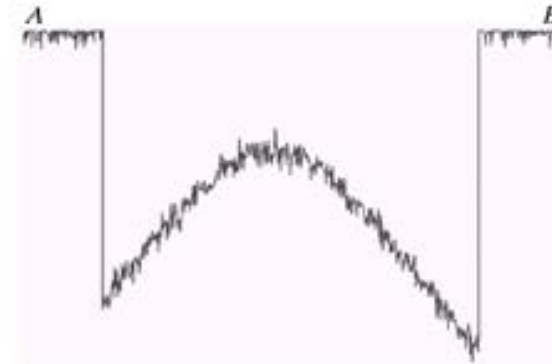
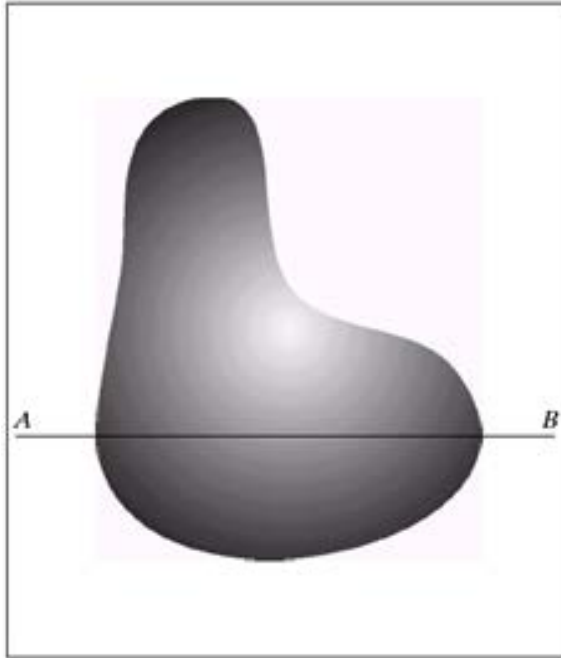
1. **Sampling** : Digitizing the coordinates values is called sampling.
2. **Quantization** : Digitizing the amplitude values is called quantization.

Image Digitization



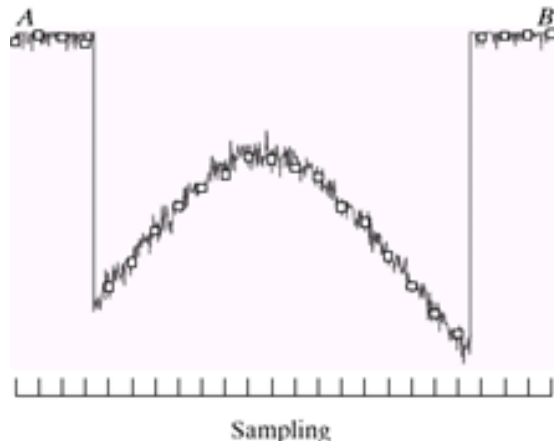
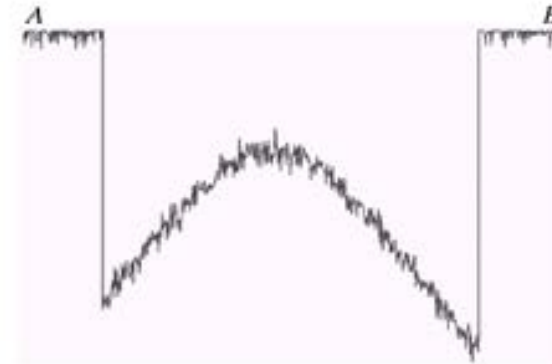
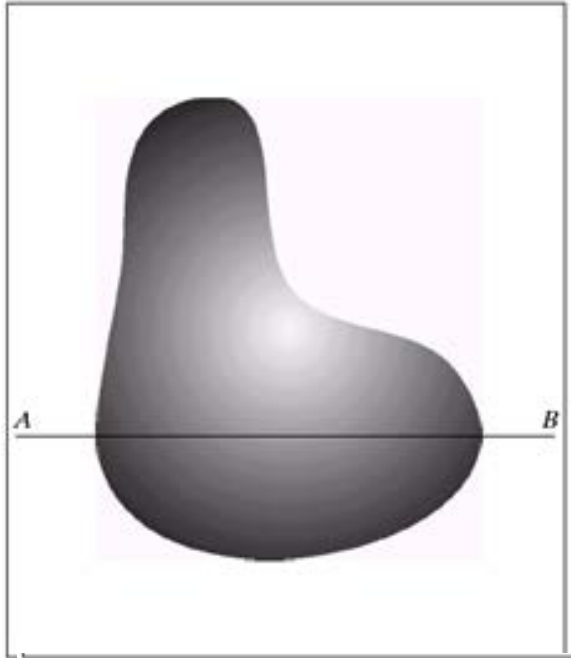
Let's start with a gray scale image.
We will try to digitize this image
along the horizontal line segment
AB

Image Digitization



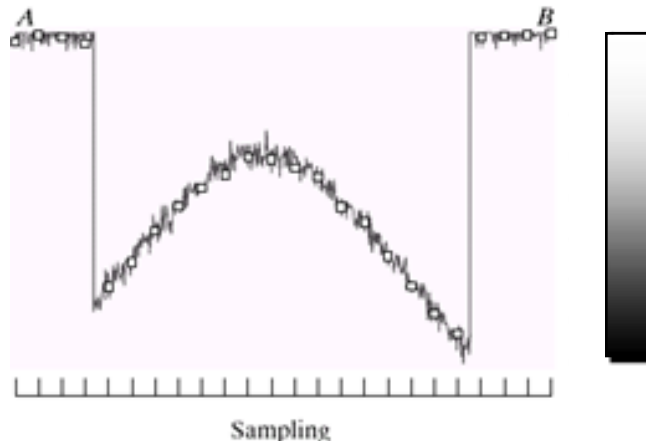
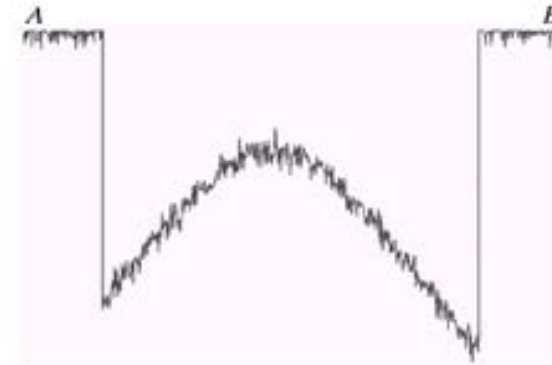
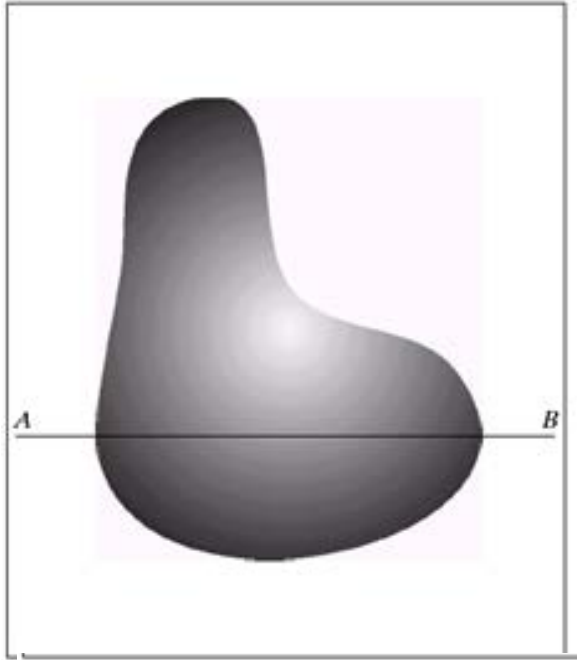
This is a plot of amplitude values of the continuous image along the line segment AB. Random variations are due to image noise.

Image Digitization



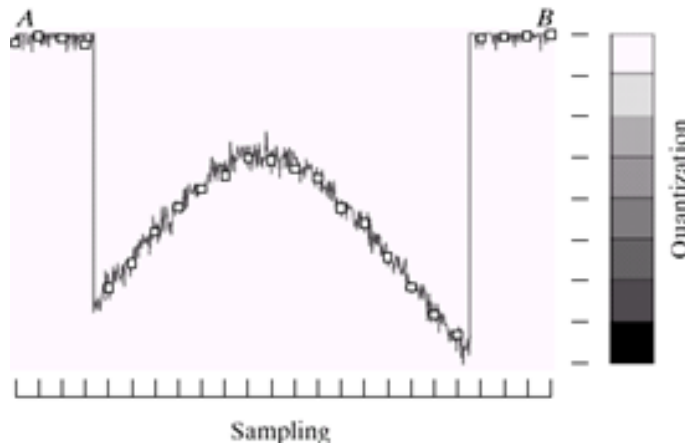
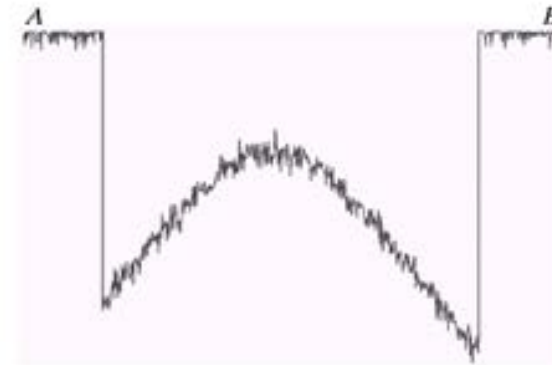
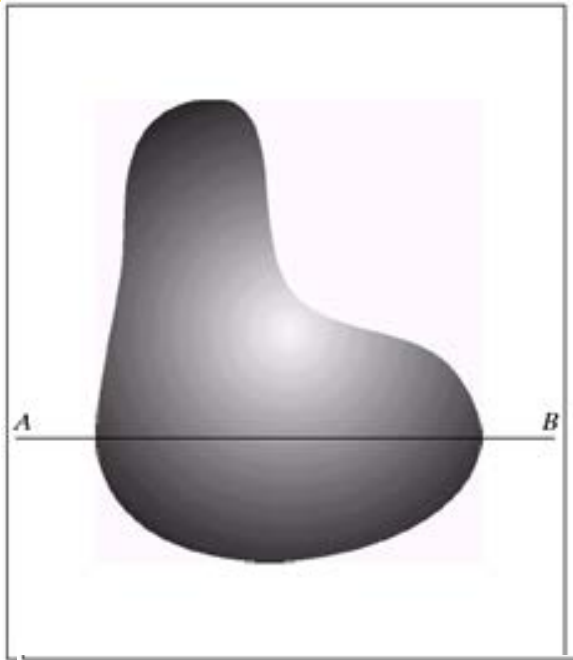
To sample this function, we take equally spaced samples along line AB

Image Digitization



The values of the samples still span a continuous range of intensity values

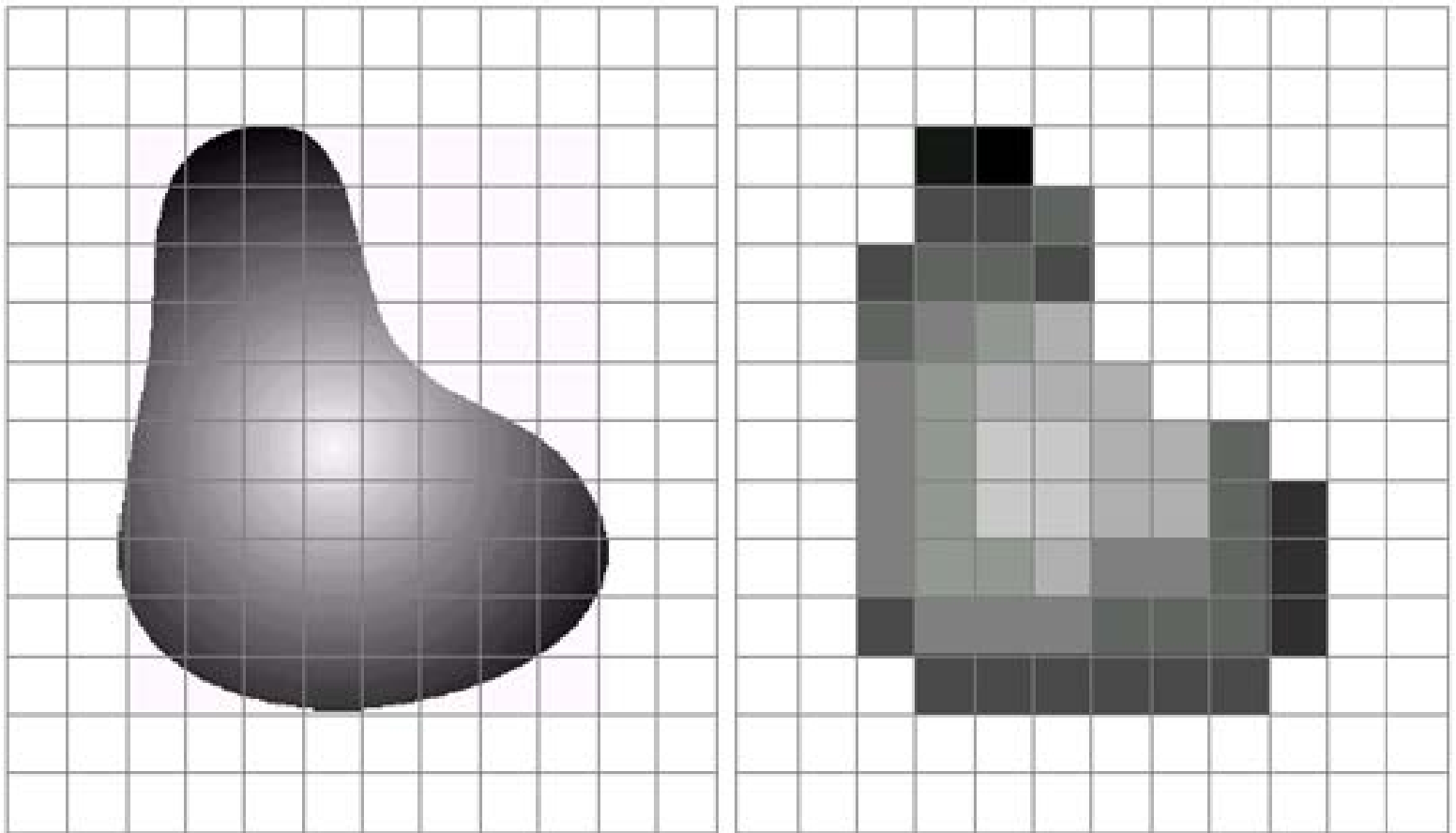
Image Digitization



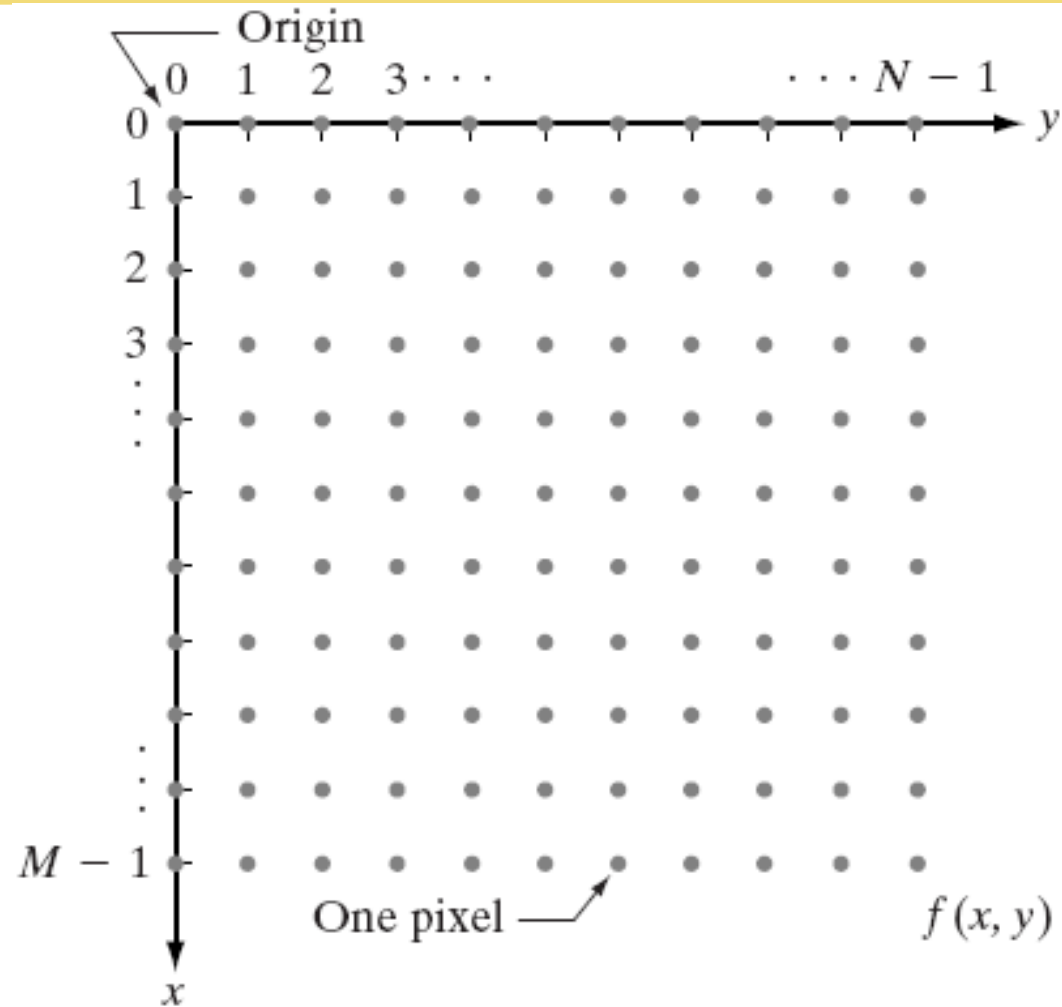
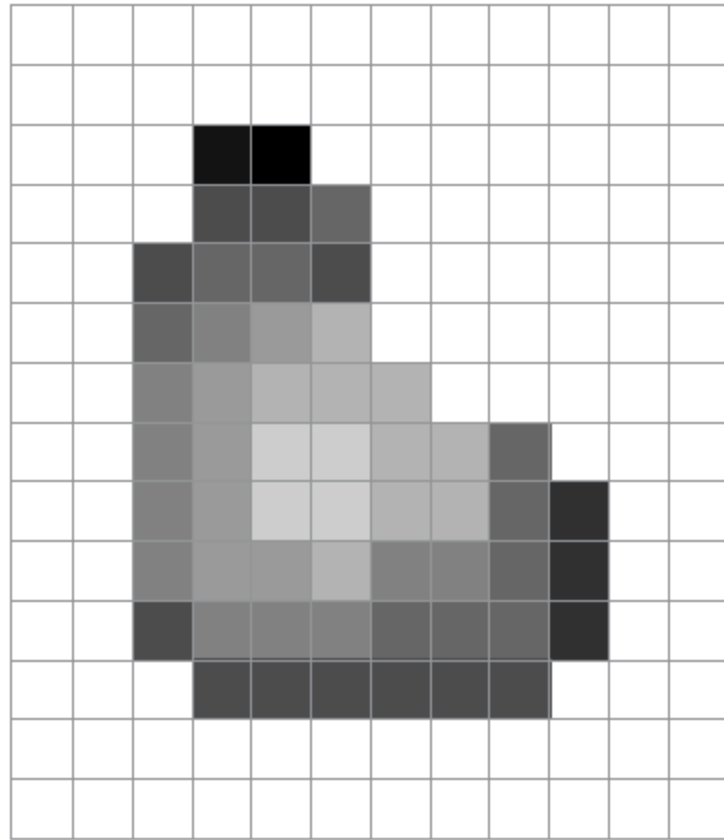
Intensity scale is divided into eight discrete intervals.

Image Digitization

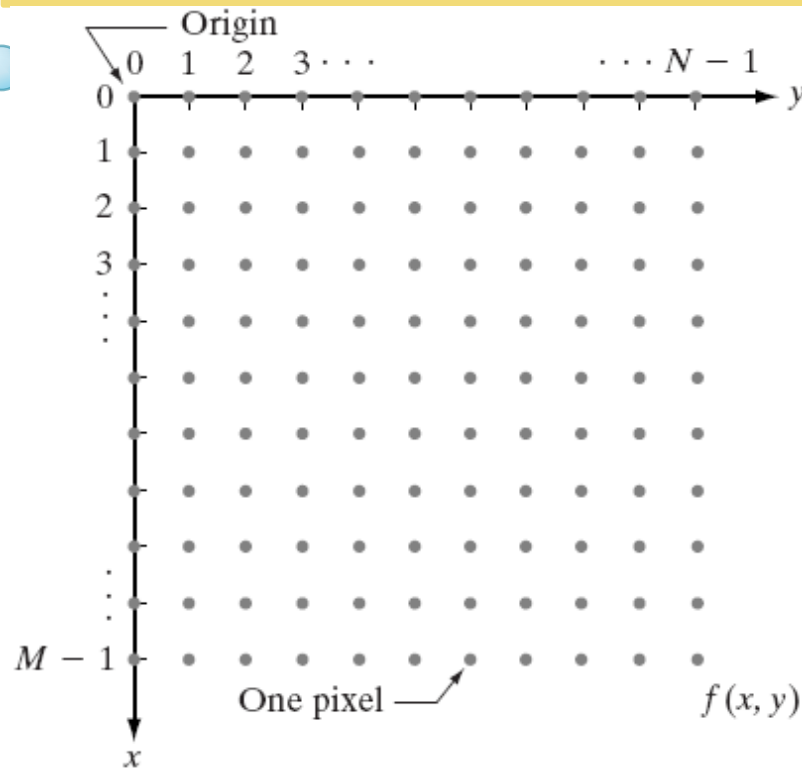
For Sensor array:



Representing Digital Images



Representing Digital Images

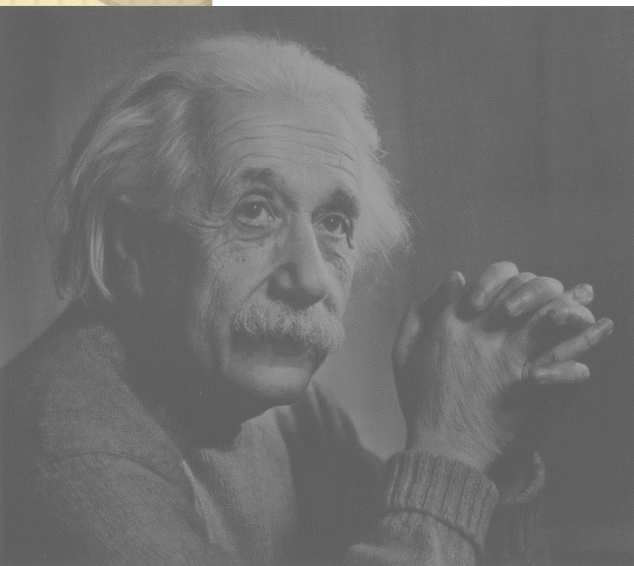


Each element of this matrix is called **pixel** or **pel**

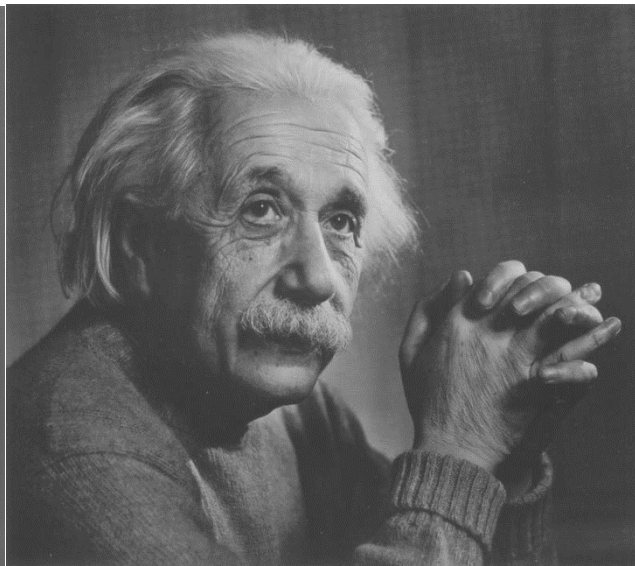
Matrix Representation

$$f(x, y) = \begin{bmatrix} f(0, 0) & f(0, 1) & \cdots & f(0, N - 1) \\ f(1, 0) & f(1, 1) & \cdots & f(1, N - 1) \\ \vdots & \vdots & & \vdots \\ f(M - 1, 0) & f(M - 1, 1) & \cdots & f(M - 1, N - 1) \end{bmatrix}$$

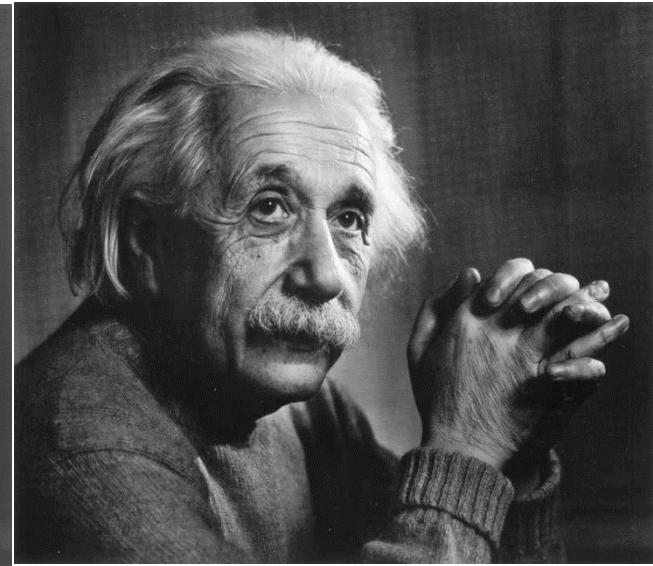
Representing Digital Images



Low contrast



Med contrast



High contrast

Representing Digital Images

- Number of Rows (M) and Columns (N) can be any integer
- How many discrete intensity levels?
 - o Number of gray levels, L, is usually power of 2:

$$L = 2^k$$

- The number of bits required to store a digitized image :

$$b = M \times N \times k$$

- For squared image :

$$b = N^2 k$$

Representing Digital Images

- Suppose, you want to create an image of size 1024×512 having 72 distinct intensity values. What will be the size of the image?

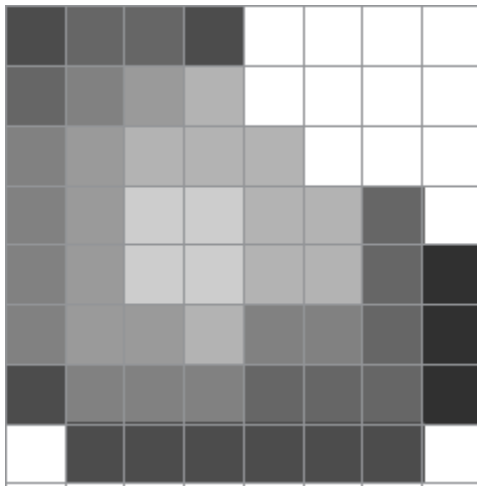


Image Interpolation

Image Interpolation

- **Interpolation** is the process of using known data to estimate values at unknown locations.
 - Zooming (Over Sampling)
 - Shrinking (Under Sampling)

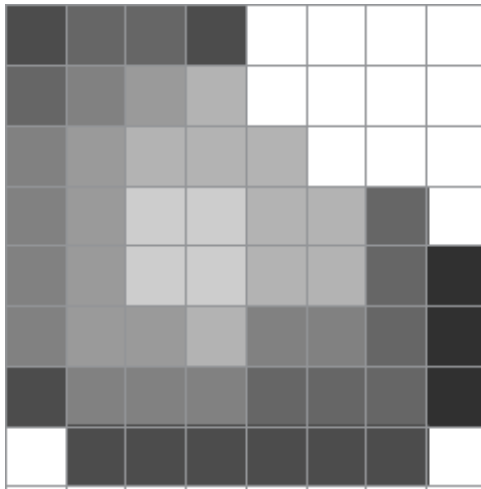
Nearest Neighbor Interpolation



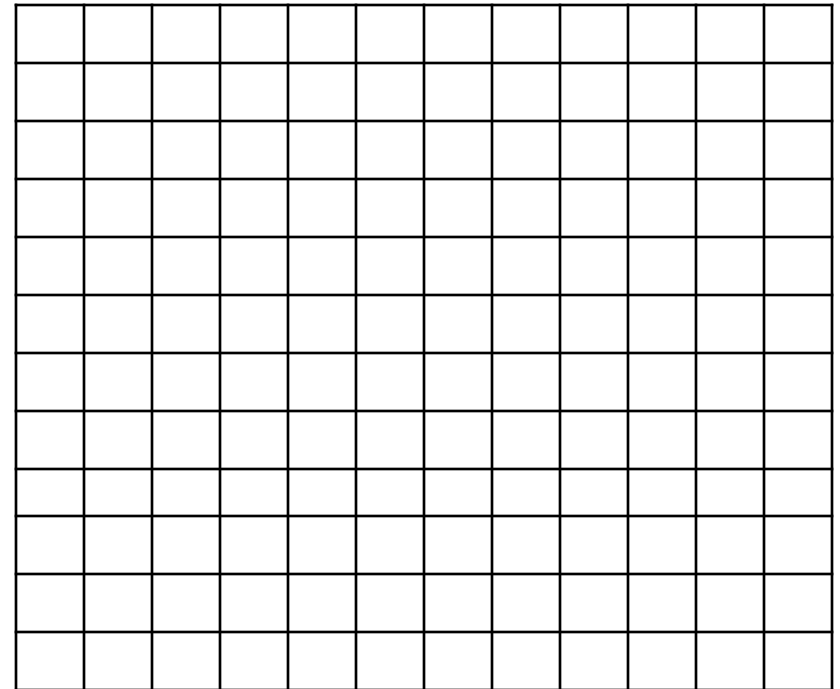
An 8X8 image
Resize this image to 12X12

Nearest Neighbor Interpolation

- ° Create an imaginary 12 X 12 grid with the same pixel spacing as the original and then shrink it so that it fits exactly over the original image.

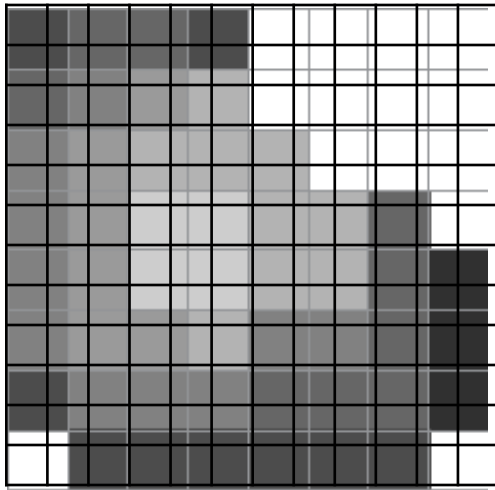


An 8X8 image
Resize this image to 12X12



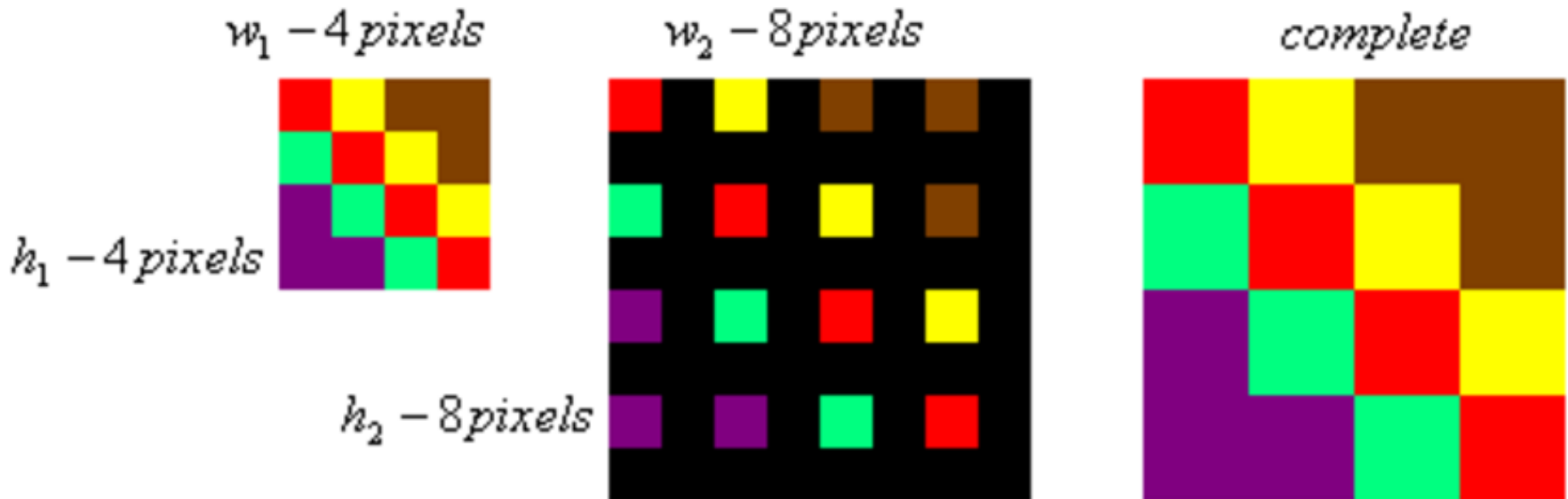
Nearest Neighbor Interpolation

- Create an imaginary 12 X 12 grid with the same pixel spacing as the original and then shrink it so that it fits exactly over the original image.



- To perform intensity-level assignment for any point in the overlay, we look for its **closest pixel** in the original image and assign the intensity of that pixel to the new pixel in the 12 X 12 grid.
- This process is also known as pixel replication method.

Nearest Neighbor Interpolation

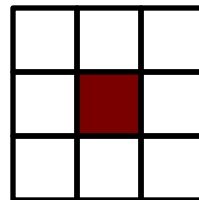




Pixel and their Relationships

Neighbors of a pixel

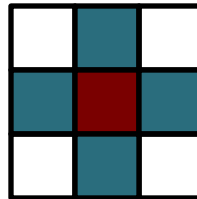
- 3 types of neighbors
 - $N_4(p)$: 4-neighbor
 - $N_D(p)$: diagonal neighbor
 - $N_8(p)$: 8-neighbor



Neighbors of a pixel

- $N_4(p)$: 4-neighbor

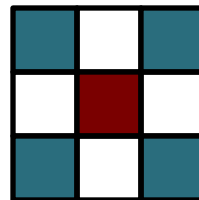
Defined as the pixels at $(x+1, y)$, $(x-1, y)$, $(x, y+1)$, $(x, y-1)$



Neighbors of a pixel

- $N_D(p)$: diagonal neighbor

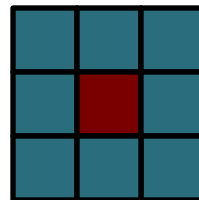
Defined as the pixels at $(x+1, y+1)$, $(x+1, y-1)$, $(x-1, y+1)$, $(x-1, y-1)$



Neighbors of a pixel

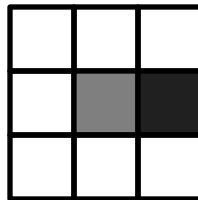
- $N_8(p)$: 8-neighbor

Defined as the pixels at $(x+1, y)$, $(x-1, y)$, $(x, y+1)$, $(x, y-1)$, $(x+1, y+1)$, $(x+1, y-1)$, $(x-1, y+1)$, $(x-1, y-1)$



Pixel Adjacency

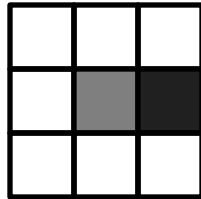
- Adjacencies depends on both
 - Neighborhood
 - Pixel gray values



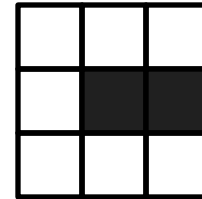
Adjacent pixels must be neighbors and have gray values from the same set, V

Pixel Adjacency

Adjacent pixels must be neighbors and have gray values from the same set, V



Can not be adjacent
if both values are
not present in V



Can be adjacent

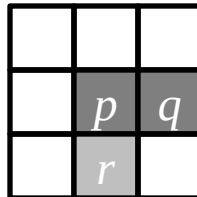
Pixel Adjacency

- 
- 3 types of adjacencies
 - 4-adjacency
 - 8-adjacency
 - m -adjacency

4-adjacency

Two pixels p , q are 4-adjacent if

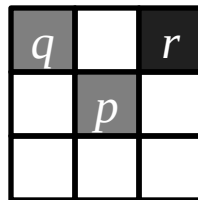
- q in $N_4(p)$ and
- p, q have values from set V



8-adjacency

Two pixels p , q are 8-adjacent if

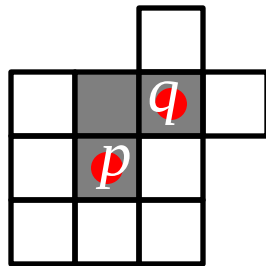
- q in $N_8(p)$ and
- p, q have values from set V



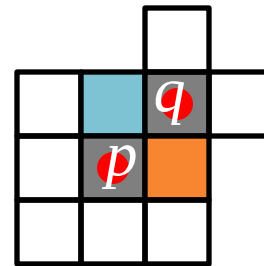
m-adjacency

Two pixels p , q are m-adjacent if

- p , q have values from set V and
 - q in $N_4(p)$, or
 - q in $N_D(p)$ and $N_4(p) \cap N_4(q)$ has no pixel with value from V



Not m-adjacent



m-adjacent



Arithmetic/Logic operations for Image Enhancement

Arithmetic/Logic operations for Image Enhancement

- Pixel by pixel operation
- Output pixel at (x, y) is calculated from input pixels of the same location, e.g., (x, y)

$$g(x, y) = f(x, y) + h(x, y)$$

$$g(x, y) = f(x, y) - h(x, y)$$

$$g(x, y) = f(x, y) \text{ and } h(x, y)$$

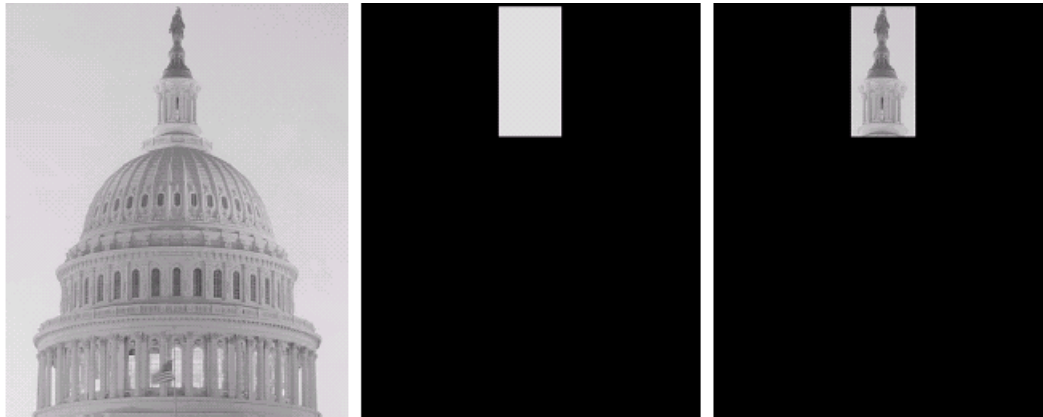
$$g(x, y) = f(x, y) \text{ or } h(x, y)$$

Arithmetic/Logic operations for Image Enhancement

- These operations are very useful for image masking.
- These operations are also useful for selecting a sub-image or region of interest.

Arithmetic/Logic operations for Image Enhancement

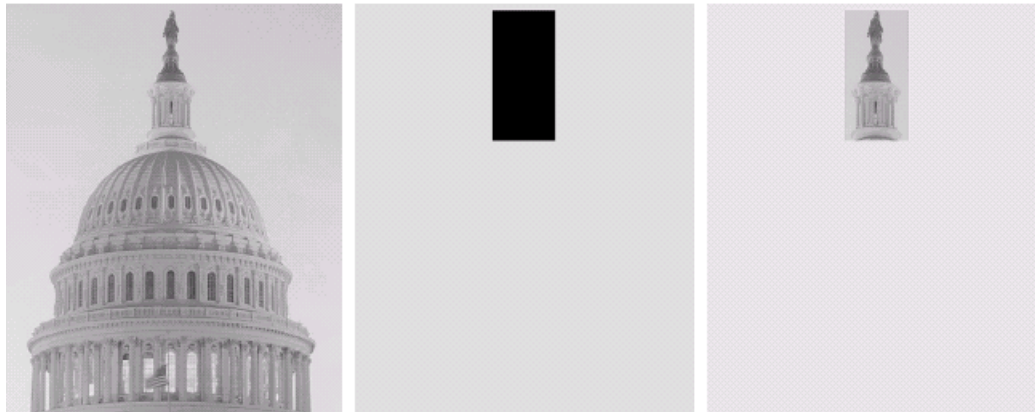
AND Masking



mask

Arithmetic/Logic operations for Image Enhancement

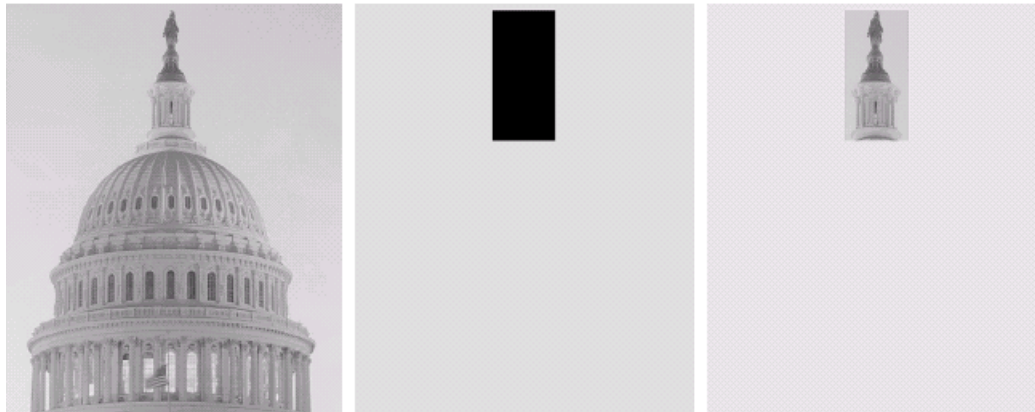
OR Masking



mask

Arithmetic/Logic operations for Image Enhancement

OR Masking



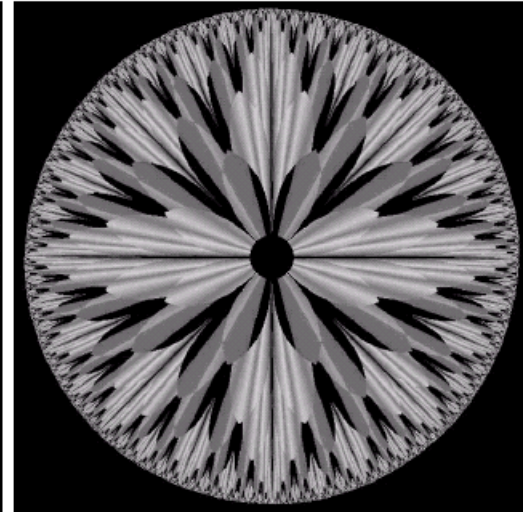
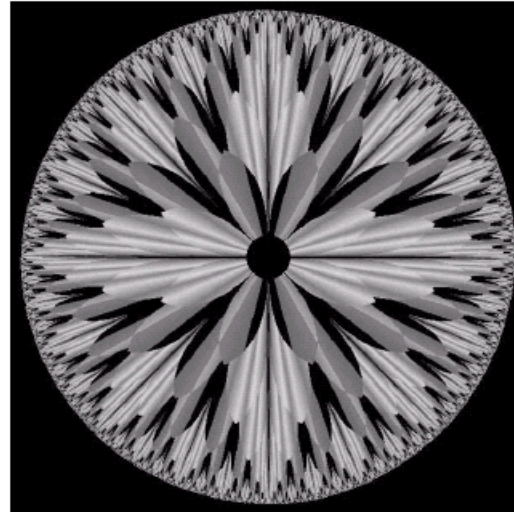
mask

Image Enhancement using Subtraction

$$g(x, y) = f(x, y) - h(x, y)$$

- Used to enhance the differences in detail
- Have commercial applications

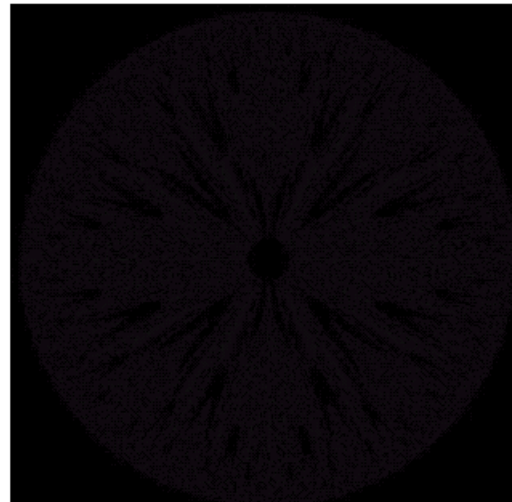
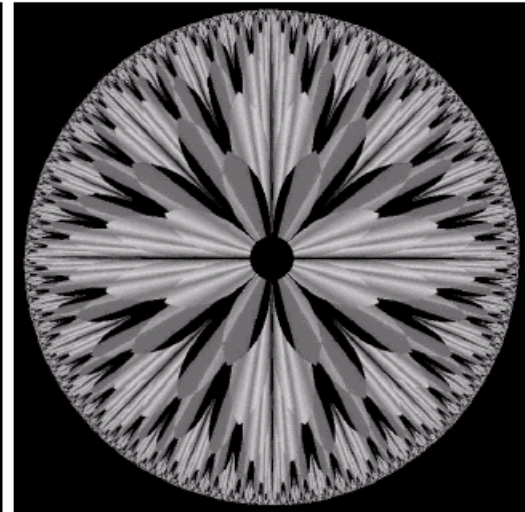
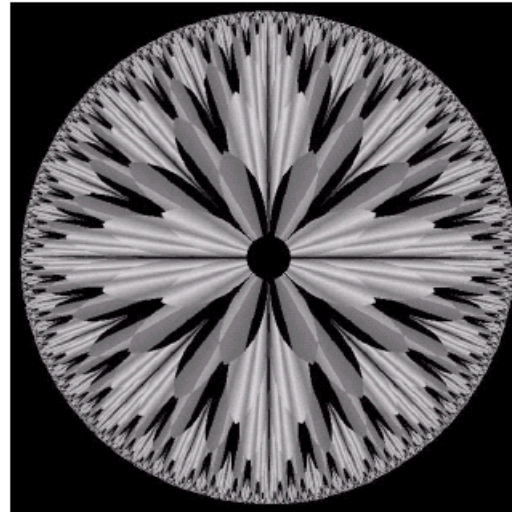
Image Enhancement using Subtraction



Least 4 bits set
to 0

Image Enhancement using Subtraction

$$g(x, y) = f(x, y) - h(x, y)$$



← Difference
image

Image Enhancement using Subtraction

$$g(x, y) = f(x, y) - h(x, y)$$

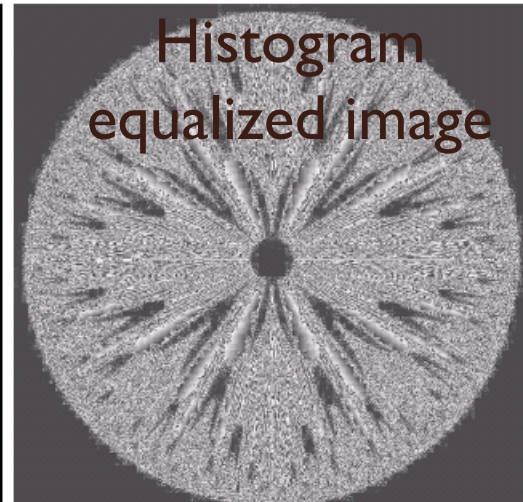
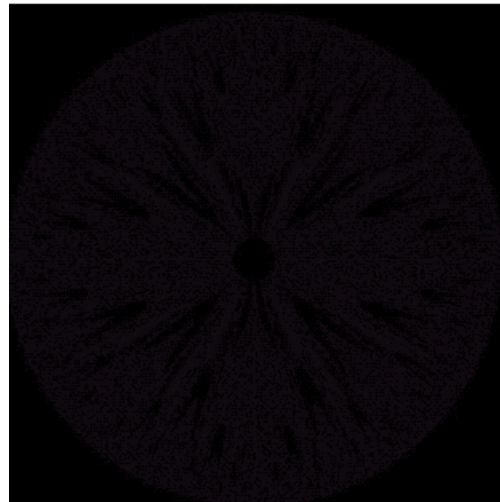
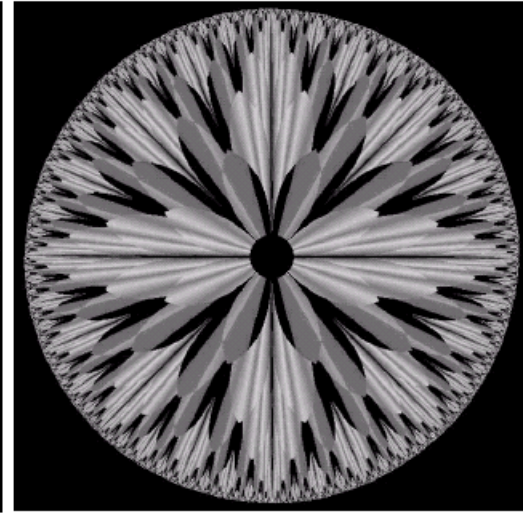
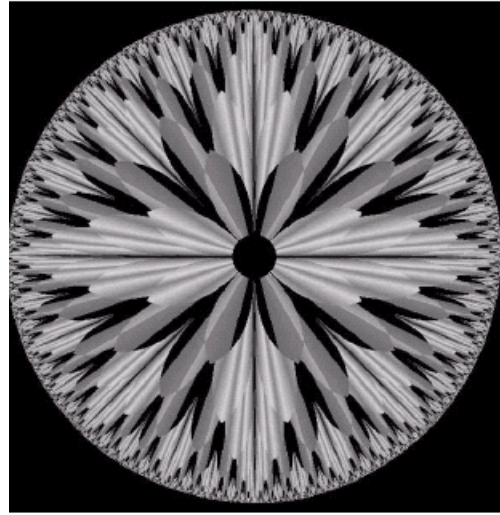
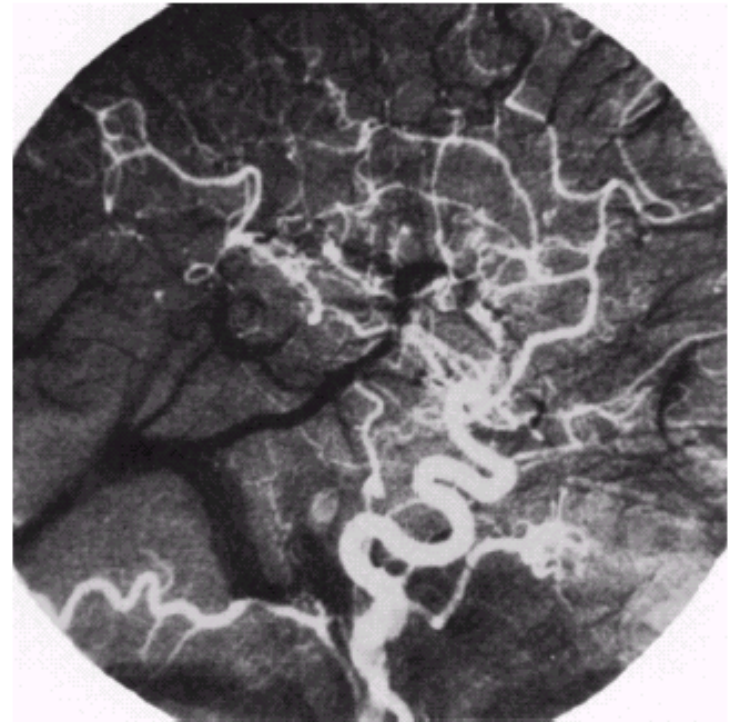
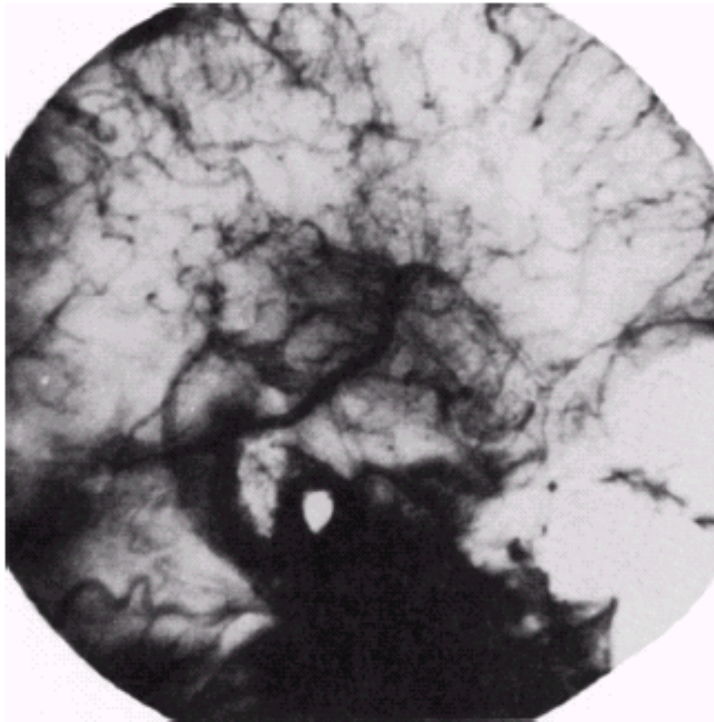


Image Enhancement using Subtraction

$$g(x, y) = f(x, y) - h(x, y)$$

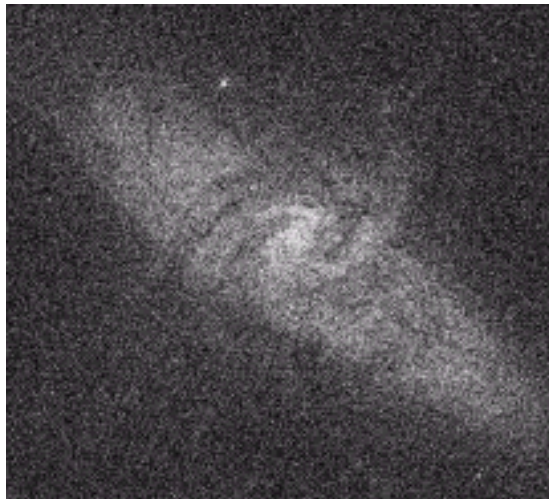
Mask mode radiography



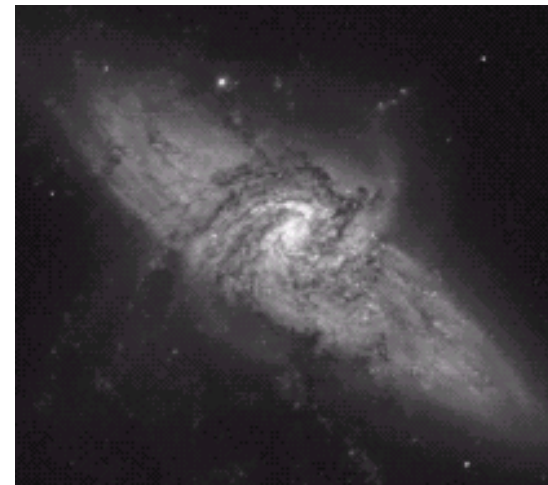
Difference Image

Image Enhancement using Addition or Averaging

- Used for noise removal



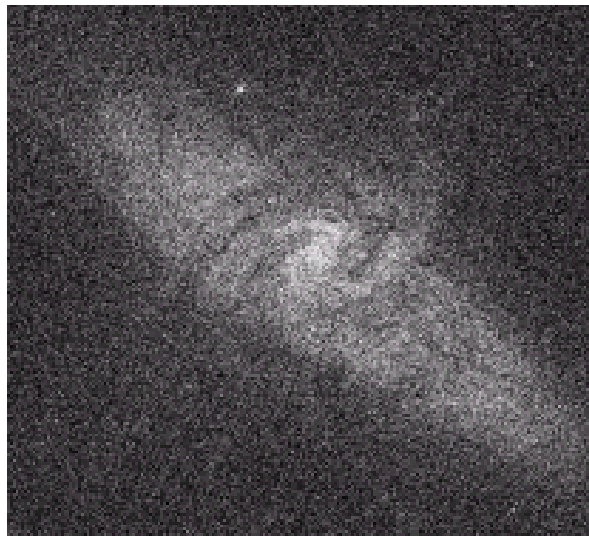
Noisy image



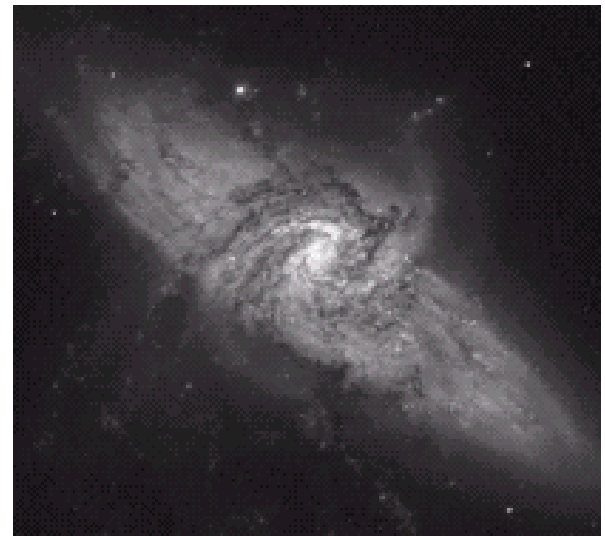
Original image

Image Enhancement using Addition or Averaging

- Used for noise removal



$g(x,y)$



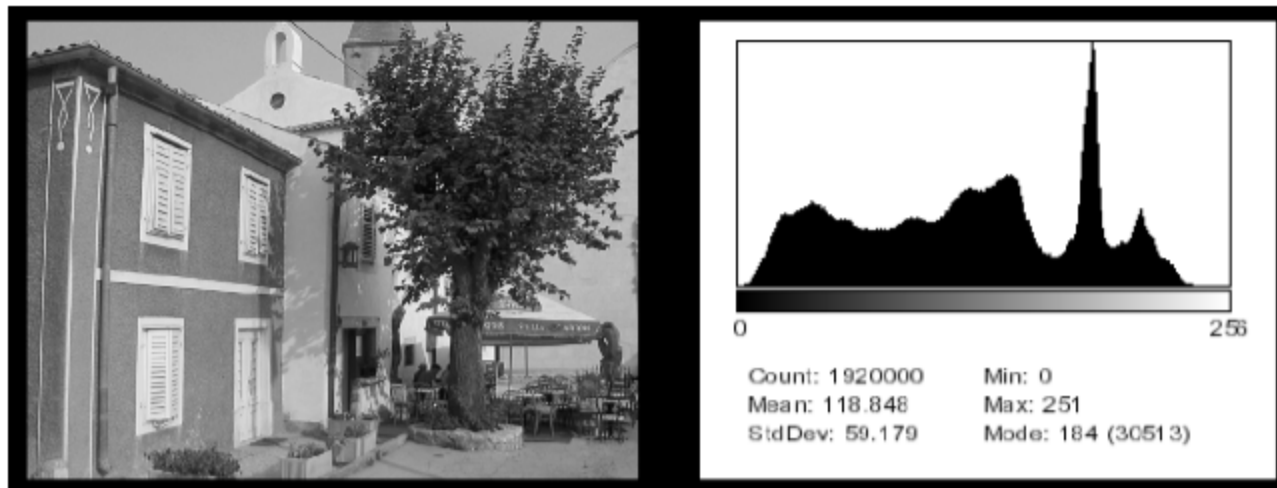
$f(x,y)$

$$g(x,y) = f(x,y) + \eta(x,y)$$

noise

Histograms

- Histograms plots how many times (frequency) each intensity value in image occurs
- Example:
 - Image (left) has 256 distinct gray levels (8 bits)
 - Histogram (right) shows frequency (how many times) each gray level occurs

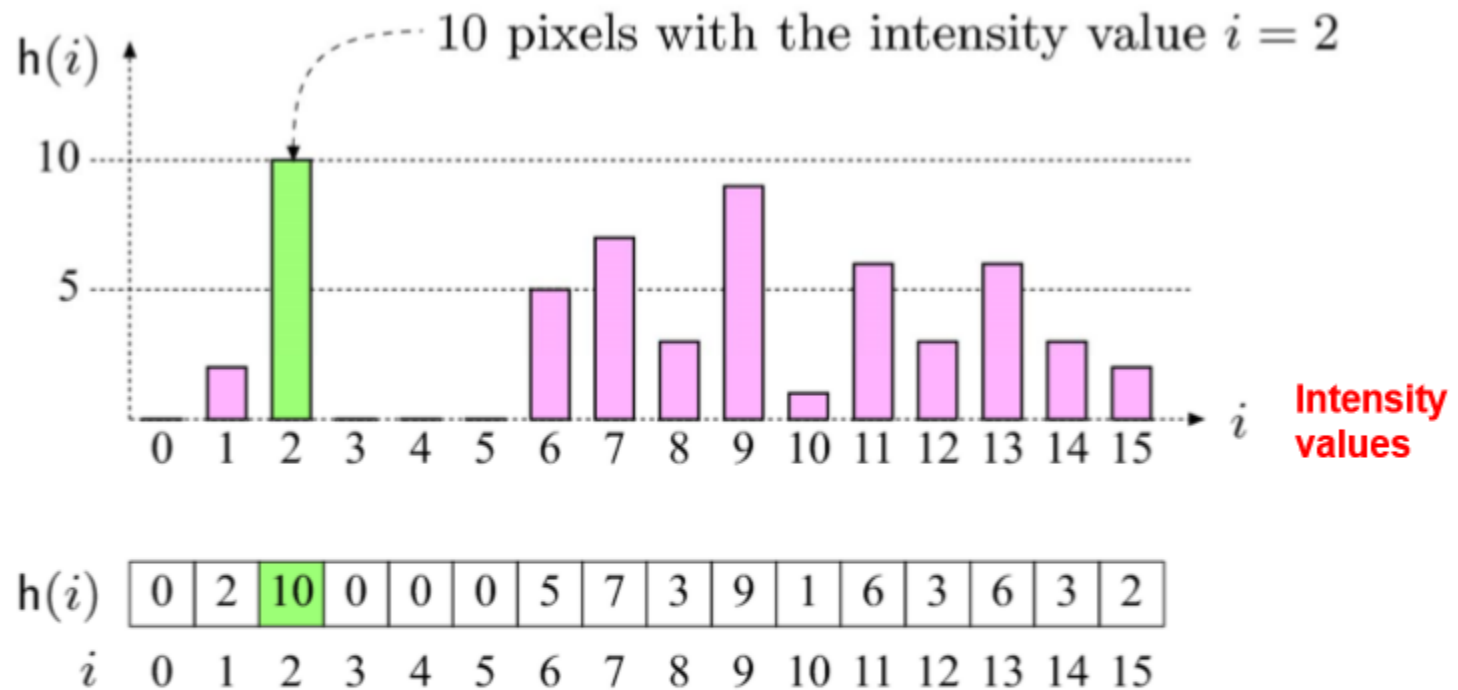


Histograms

- Many cameras display real time histograms of scene
- Helps avoid taking over-exposed pictures
- Also easier to detect types of processing previously applied to image



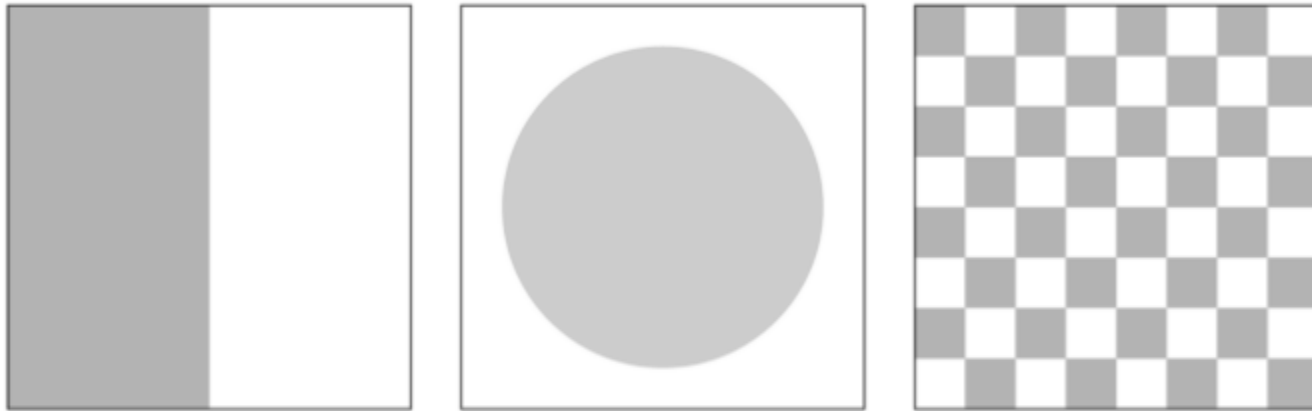
Histograms



- E.g. $K = 16$, 10 pixels have intensity value = 2
- Histograms: only statistical information
- No indication of **location** of pixels

Histograms

- Different images can have **same** histogram
- 3 images below have same histogram



- Half of pixels are gray, half are white
 - Same histogram = same statistics
 - Distribution of intensities could be different
- Can we reconstruct image from histogram? No!

Histograms

- So, a histogram for a grayscale image with intensity values in range

$$I(u, v) \in [0, K - 1]$$

would contain exactly K entries

- E.g. 8-bit grayscale image, $K = 2^8 = 256$
- Each histogram entry is defined as:

$h(i)$ = number of pixels with intensity i for all $0 \leq i < K$.

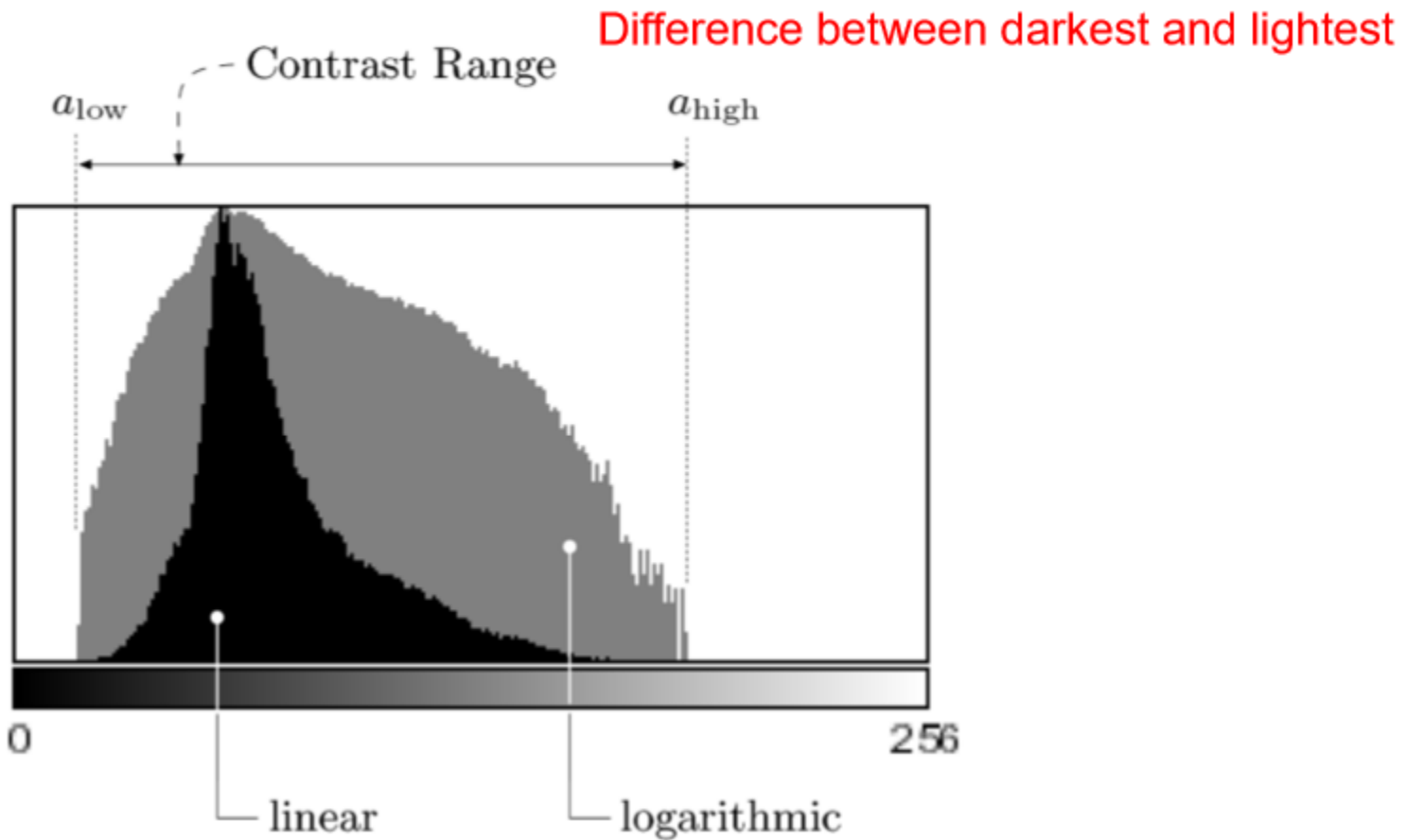
- E.g: $h(255)$ = number of pixels with intensity = 255
- Formal definition

$$h(i) = \text{card}\{(u, v) \mid I(u, v) = i\}$$

Number (size of set) of pixels

such that

Histograms



Histograms

- Histograms help detect image acquisition issues
- Problems with image can be identified on histogram
 - Over and under exposure
 - Brightness
 - Contrast
 - Dynamic Range
- Point operations can be used to alter histogram. E.g
 - Addition
 - Multiplication
 - Exp and Log
 - Intensity Windowing (Contrast Modification)

Histograms

- Brightness of a grayscale image is the **average intensity** of all pixels in image

$$B(I) = \frac{1}{wh} \sum_{v=1}^h \sum_{u=1}^w I(u, v)$$

2. Divide by total number of pixels

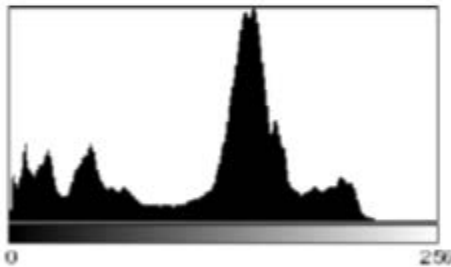
1. Sum up all pixel intensities

Histograms

Exposure? Are intensity values spread **(good)** out or bunched up **(bad)**

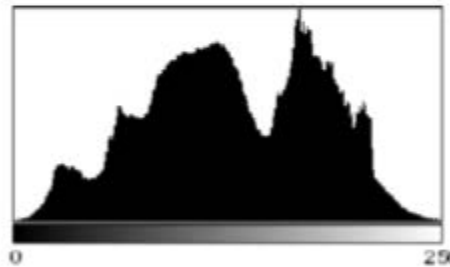


Image



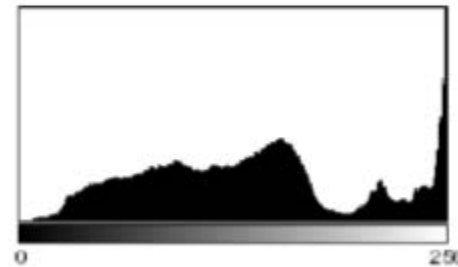
(a)

Underexposed



(b)

**Properly
Exposed**



(c)

Overexposed

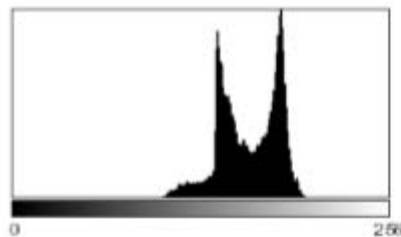
Histogram

Histograms

- The contrast of a grayscale image indicates how easily objects in the image can be distinguished
- **High contrast image:** many distinct intensity values
- **Low contrast:** image uses few intensity values

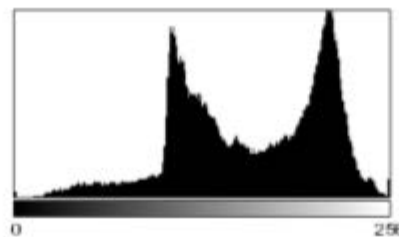
Histograms

Good Contrast? Widely spread intensity values
+ large difference between min and max intensity values



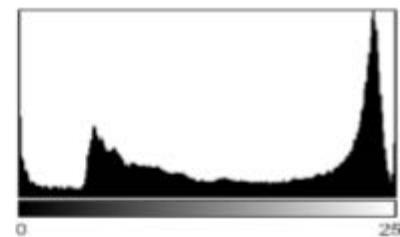
(a)

Low contrast



(b)

Normal contrast

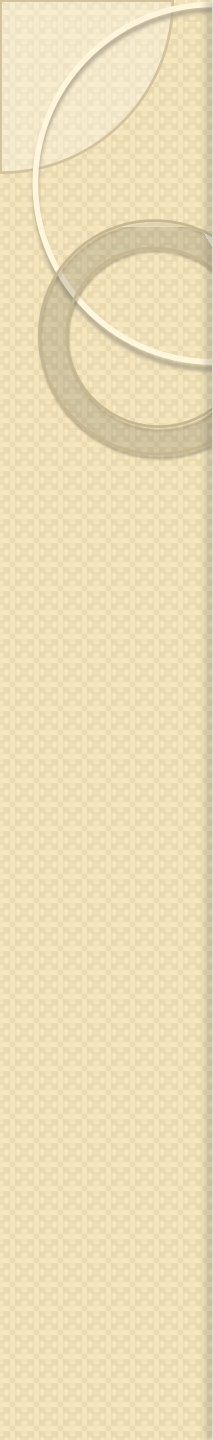


(c)

High contrast

Image

Histogram



THANK YOU!!!